



College of Information Technology and Engineering
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“Techno-Fitness: Leveraging Computer Vision and Workout Assistance Coach.”

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ABSTRACT

Many beginners struggle with maintaining a fitness routine due to time constraints, lack of knowledge, courage, and difficulty in finding professional assistance. Fitness applications often lack the capabilities to monitor and assist the user performing the exercise in real time. With this, the Techno-Fitness Device aims to address this issue. It is a computer vision (OpenCV) powered device that utilizes Mediapipe for the detection of exercise form designed to improve exercise consistency, provide workout assistance, and engage the user in doing exercises. This research shall explore and evaluate how the integration of OpenCV can make way for the Mediapipe framework to detect the overall posture and form of an exercise being performed by an individual. Its target audience are beginner to intermediate fitness level users and calisthenics or home-based workout enthusiasts. The device utilizes integrated circuits partnered with software to detect, display, notify, and execute the desired functions of the project. Built on a Raspberry Pi 5, the components featured in the device are a camera, elevation motor, monitor, and biometrics to allow an optimum performance of the device. Results of the study show that the device is able to detect posture of an exercise, be able to count reps, and provide feedback through both visual and auditory cues to ensure the user is performing a workout. Testing revealed that the device delivers optimum performance in posture recognition within optimal lighting and distance conditions, reinforcing its reliability for home-based calisthenics training. Overall, the study concludes that the Techno-Fitness Device presents a viable, innovative, and efficient device for assisting users when it comes to improving workout consistency, promoting safe exercise habits, and fostering motivation among beginner to intermediate fitness enthusiasts.

Keywords – Techno-Fitness, Artificial Intelligence, OpenCV, Mediapipe, Fitness, Consistency, Pose Detection, Injury Prevention, Raspberry Pi 5

INTRODUCTION

Background of Project

The COVID-19 pandemic significantly reshaped public awareness of health and fitness, highlighting the essential role of physical activity in maintaining both physical and mental well-being. This shift led to a notable rise in home-based workout routines, with calisthenics emerging as one of the most accessible and preferred methods. Calisthenics involves bodyweight exercises such as squats, push-ups, and planks, which require little to no equipment. This makes them especially suitable for individuals exercising at home or in limited spaces. The growing interest in this form of training has been fueled by increased health consciousness, the need for flexible routines, and the widespread availability of digital workout content. According to Harvard Health (2021), calisthenics offers a cost-effective and efficient way to build strength, flexibility, and endurance without relying on gym facilities or complex tools. Moreover, Vogue (2020) notes that younger generations are increasingly drawn to calisthenics not only for physical appearance but also for developing movement control and functional fitness.

Numerous studies support the effectiveness of home-based calisthenics programs, including for individuals with specific medical conditions. For example, Altan et al. (2016) demonstrated that patients with Ankylosing Spondylitis who followed a home-based calisthenics program experienced significant improvements in mobility and quality of life. Similarly, Karapolat et al. (2014) found that individuals with Multiple Sclerosis showed enhancements in balance and coordination after engaging in structured bodyweight exercises. These findings affirm the adaptability and therapeutic potential of calisthenics for a broad range of users.

Despite these advantages, a persistent challenge in home-based training is the lack of real-time feedback on posture and form, which are critical for exercise safety and effectiveness. Most individuals, particularly beginners and those at the intermediate level, rely heavily on mobile applications for guidance. However, several studies have identified limitations in these fitness apps. A systematic review by Direito et al. (2017) found that many applications lacked adherence to established exercise science principles and failed to provide dynamic feedback. Similarly, Middelweerd et al. (2014) reported that most physical activity apps offered limited behavioral support and inadequate instructional content.

This gap presents a significant risk, especially in calisthenics, where precision in form is vital to prevent injury and ensure progress. Without expert supervision or responsive assistance, users may unknowingly perform exercises incorrectly, reducing their effectiveness and increasing the likelihood of strain or injury. To address this issue, the project proposes the development of a computer vision based fitness system utilizing OpenCV and Mediapipe. The two work hand in hand into providing pose detection. OpenCV is an open-source software library designed for computer vision, image processing, and machine learning. It allows computers to “see” and interpret visual data from sources like cameras, videos, or images. Meanwhile, MediaPipe helps computers understand what’s happening in an image or video, such as identifying body movements, facial expressions, or hand gestures all in real time. MediaPipe acts as the system’s “posture analyst” as it identifies and tracks the user’s body landmarks through the camera feed. By visually tracking posture and movement in real time, the system aims to assist users in performing exercises with proper form. It is specifically designed for beginner to intermediate users and will include tailored workout variations to match their skill levels. This responsive approach bridges the gap between accessibility and professional oversight, enhancing the safety, effectiveness, and overall quality of home-based calisthenics training. With this, The Techno-Fitness device is a computer vision (OpenCV) and MediaPipe powered device, and an interactive training assistant designed to help users perform exercises with correct posture and form. Using a camera connected to a Raspberry Pi 5, the device detects key body landmarks and calculates joint angles, which are then compared against predefined references and motion rules. Exercises are divided into three categories: *static*, *repetitions*, and *dynamic*. Each type has their own way of detecting exercise posture however they fundamentally have a reference model or motion rule to be compared to one another if they are accurate using angles. Based on this comparison, the system determines whether the user’s posture aligns with the ideal form. If discrepancies are detected, the device provides immediate feedback through a combination of audio cues (such as alerts and tunes) and visual indicators (on-screen rep counter and matching percentage). These cues help the user to determine if they are doing the wrong posture.

To enhance user motivation and long-term engagement, Techno-Fitness incorporates gamification elements, such as progress tracking, live feed of the detection, and performance scores. This not only makes the experience enjoyable but also encourages consistent

improvement over time. By integrating such, the Techno-Fitness device offers a smart, safe, and effective way for individuals to improve their workout techniques and achieve better fitness outcomes.

Statement of the Problem

Despite increasing public awareness of the importance of health and fitness, many individuals, particularly beginners and intermediate calisthenics practitioners, struggle to perform exercises with proper form and consistency. This is largely due to the lack of real-time assistance in most fitness applications. While mobile fitness apps offer convenience and accessibility, studies have shown that they often fail to provide accurate, adaptive instructions tailored to individual performance (Direito et al., 2017; Middelweerd et al., 2014). As a result, users may unknowingly perform movements incorrectly, increasing the risk of injury and limiting the effectiveness of their workouts.

Although hiring a professional fitness coach could mitigate these issues, it is often cost-prohibitive and difficult to schedule consistently (Kotte et al., 2023). This leaves a significant gap in the market for a solution that is both affordable and capable of offering intelligent, real-time feedback. Moreover, many beginners struggle to maintain a consistent and motivating routine, especially when faced with exercise challenges that do not match their fitness level. Existing platforms rarely adapt workouts to user progression, further contributing to disengagement and improper technique. Therefore, there is a critical need for an accessible, adaptive, and cost-effective fitness solution that not only offers progressive workout variations but also ensures proper exercise form through real-time posture analysis, bridging the gap between professional guidance and digital convenience.

Objectives

This research is motivated by the urgent need to improve the user engagement, workout accuracy and long-term consistency in personal fitness. Despite the rapid growth of digital fitness tools, there is still a shortage of solutions that successfully integrate real-time feedback, user-specific instruction, and motivating reinforcement. The important goal of this study is to develop and evaluate a fitness device that uses computer vision to detect correct posture of an



exercise and to provide an engaging user experience to enhance their fitness journey. Furthermore, the project supports two United Nations Sustainable Development Goals: SDG #3 – Good Health and Well-Being, by promoting accessible fitness solutions that encourage healthier lifestyles, and SDG #4 – Quality Education, by fostering self-learning in health and physical fitness through real-time workout assistance. The device aims to create a system that utilizes OpenCV and Mediapipe to analyze posture and movements while also providing a quick feedback such as visual/auditory cues that can help the user to understand more on performing their workouts. Moreover, the study also hopes to achieve several primary objectives. The objectives are as follows:

General Objective:

Develop a prototype of a device that utilizes real-time body keypoint detection using OpenCV to monitor, analyze, and correct users' posture during physical exercises.

Specific Objectives:

1. To explore and test the posture detection capabilities of a pose detection API by incorporating first simple exercises into the system.
2. To apply the posture detection feedback utilizing pose detection API to each exercise in the workout plan using a microcontroller.
3. To implement a feature utilizing pose detection API that compares the posture of the users to the predefined and coded correct postures listed in the calisthenics workout plan of the developers.
4. To integrate a biometric authentication system, specifically fingerprint recognition, to provide secure and multiple user access to the device.
5. To integrate a camera elevation system to accommodate both ground based and standing based workouts.
6. To integrate light sensors to detect luminosity of the room to ensure the API's accuracy.
7. To evaluate the system's efficiency and effectiveness in delivering real-time visual and auditory cues to users through posture detection using body points, assessing consistency of detection within the system's operational limits.

In the long run, the study hopes to contribute to the field of fitness technology by offering an effective solution that not only tracks progress but also encourages long-term health behaviours. This study establishes a link between sophisticated technology capability and the psychological components of sustaining a healthy lifestyle. A connection that is undeveloped in existing fitness systems.

Significance of the Study

The Techno-Fitness project tries to address a growing demand for accessible, guided, and cost-effective fitness solutions in the context of increasing health consciousness and a shift toward home-based workouts. As highlighted in existing literature, calisthenics proved to be an effective ideal fitness approach for individuals without access to gym equipment or facilities. However, a critical issue persists: most mobile fitness applications lack real-time assistance during exercise execution, posing risks of improper execution of exercises, which can lead to ineffective results or injury. The only reliable sources of real-time correction currently available are professional fitness coaches and vision-based systems like Techno-Fitness.

While professional trainers can provide individualized guidance, they are often financially out of reach for many individuals. A local cost assessment of fitness facilities in Lipa City revealed that gym memberships can cost as much as ₱30,000 annually, with personal training sessions priced between ₱3,000 and ₱3,200 for 8 to 10 sessions. Some gyms charge ₱650 per session, and even low-cost gyms charging ₱100 per visit provide minimal personalized instruction. These financial and social barriers limit access to structured physical activity for many potential users.

To determine the market demand and relevance of the Techno-Fitness system, a survey was conducted with 32 purposely sampled individuals aged 18 to 45, including students, professionals, and casual exercisers. The majority of respondents were male (65.6%), and the largest age group was 21 to 23 years old (53.13%). In terms of fitness level, 50% identified as intermediate and 40.6% as beginners, which aligns with the system's target user base.

Survey results reveal important insights: while participants generally agreed that they are engaged in fitness activities (mean = 3.41) and are familiar with different workout types (mean =

3.53), they expressed neutrality regarding consistency in workouts (mean = 3.38) and challenges with maintaining correct form (mean = 3.25). These findings underscore a common problem among self-guided exercisers: difficulty in maintaining proper technique and long-term discipline. Importantly, the highest-rated response came from the question regarding interest in an AI-powered fitness device that offers real-time movement feedback, which received a mean score of 4.22 (Strongly Agree). This indicates a strong interest in a solution like Techno-Fitness that can bridge the gap between fitness app limitations and expensive coaching.

By utilizing OpenCV and Mediapipe on a Raspberry Pi platform, Techno-Fitness simulates the feedback traditionally provided by a fitness coach. It offers posture correction, and guided routines in real time. This not only enhances the safety and effectiveness of home workouts but also fosters user autonomy and motivation. Ultimately, the Techno-Fitness project contributes to the development of inclusive, technology-driven health and wellness tools that respond directly to the needs and preferences of modern fitness enthusiasts.

Scope and Limitation

This study focused on the development of the **Techno-Fitness device**, a hardware-based fitness assistance system that integrates computer vision to provide real-time posture assistance during calisthenics workouts. The system is specifically designed for individuals engaging in home-based bodyweight training and addresses common challenges found in fitness applications, such as the lack of real-time assistance. By leveraging **OpenCV**, an open-source computer vision library, the device enables the **MediaPipe** library to detect the form of the user whilst executing an exercise. It visually monitors user movement through a camera feed and delivers assistance through the use of audio and visual cues such as displaying matching percentage, counting of reps and constantly monitoring the posture being done. This section outlines the scope and limitations of the project.

Scope of the Study

- **Target User Group: Beginners to Intermediate Calisthenics Practitioners** - The device is tailored for individuals with limited to moderate experience in calisthenics. These users typically benefit from structured assistance, form correction, and progressive



training plans, which the system provides.

- **Support for Home-Based Workouts** - Techno-Fitness is developed and designed for users who prefer or require exercising at home. It specializes in beginner-to-intermediate calisthenics. By eliminating the need for specialized equipment, it promotes convenient, self-guided fitness.
- **Calisthenics-Focused Workout Programming** - All exercises integrated into the device adhere to calisthenics principles, relying solely on bodyweight movements. This allowed users to perform safe and effective full-body workouts without external weights or machines.
- **Integration of OpenCV and Mediapipe for Real-Time Posture Detection** - The device successfully utilized the mentioned libraries to track and analyze body posture and form in real time through a camera feed. This computer vision system enables visual feedback and audio cues (tunes, sound effects) that helps users monitor their form, improve workout performance and reduce injury risk.
- **Posture Analysis via Angle Measurement** - Each exercise included reference angles derived from verified reference media and motion rules. The system compares user posture against these angles to evaluate and checks the movement in real-time.
- **Workout Readiness Verification System** - In the implemented *Prerequisite Tab*, it ensures that all conditions are met before starting a workout session. Before proceeding with a workout, users were prompted to meet certain prerequisites, such as the camera working, at least 80% of the body landmarks are visible within the frame and lighting conditions were adequate. Once these conditions were satisfied, the user would now be directed to the proper workout session.
- **Fixed Workout Plans Validated by a Fitness Coach** - The exercise routines are pre-designed by the researchers and validated by a certified fitness professional to ensure

safety, effectiveness, and appropriate progression for the target user level. Feedback from the validation confirmed the appropriateness and effectiveness of the exercise plans.

- **Utilization of JSON files for data storing** - The device utilized JSON files for storing account credentials and performance statistics. This was used because it offers a straight-forward solution for data storing as using a database would put additional strain onto the already work intensive pose detection.
- **Utilization of an Arduino Interface for Fingerprint and Light Detection** - An arduino interface is responsible for managing specific peripheral hardware, namely the fingerprint module to provide secure, biometric user authentication and the digital light dependent resistor (light sensor) to detect ambient luminosity, ensuring optimal lighting conditions for the API's detection accuracy. The Arduino gathers data from these sensors and communicates it to the central Raspberry Pi 5 for processing.
- **Individual User Accounts and Progress Tracking** - The device supports multiple users by maintaining individual user accounts utilizing JSON files. Each user's progress is tracked independently.
- **Camera Elevation Mechanism to cater both Standing based and Ground based exercises** - The integration of a 12V linear actuator provides a reliable way to adjust the camera's height, allowing the system to capture both standing-based and ground-based exercises efficiently. This increased the range of supported movements and improved user experience during trials.
- **Grade/Performance-Based Workout Scheduler** - The device features a dynamic scheduler wherein it determines what schedule would the user undertake every day and week. Certain conditions are implemented which were validated by the fitness coach. This schedule is strictly monitored as there are parameters users must meet in order to progress to the workout.



- **Offline Hardware-Based System** - Techno-Fitness functions operates entirely offline without relying on cloud infrastructure or mobile applications, making it ideal for use in environments with limited internet connectivity.

Limitations of the Study

- **No AI-Generated Workout Content** - Although OpenCV and Mediapipe were utilized for computer vision and posture detection, there was no integration of an AI system that generates personalized workout plans. All routines are manually developed, tested and externally validated by a fitness professional.
- **Simple Effective Exercises are Selected for the workout** - Due to the current processing limitations of the Raspberry Pi 5, the system is limited to a set of simple controlled exercises that are compatible with its real-time detection capabilities. Exercises involving explosive or high-velocity movements are excluded to maintain accuracy and form analysis.
- **No Cloud or Mobile App Integration** - The prototype operates offline and does not support synchronization with mobile apps or cloud-based fitness platforms, which limits long-term data storage and remote access.
- **Basic User Interface via Gyro Mouse and Keyboard** - User interaction is restricted to a gyro mouse and keyboard. Touchscreen and voice controls were not implemented in the current prototype.
- **No Real-Time Dynamic Content Personalization** - The system did not dynamically adapt workouts based on individual user progress in real time beyond the preset progression built into the routines.



- **Mediapipe Pose Detection Limitations** - Mediapipe based detection requires a well-lit environment for accurate posture recognition. In addition, the pose detection works depending on the perspective of the camera, and partial occlusion of certain body parts to which the framework won't be able to detect or process.
- **Mediapipe Pose Detection Limitations** - MediaPipe Pose's accuracy is highly dependent on environmental conditions. It requires a bright, well-lit environment and is sensitive to occlusion, where the model sometimes could not detect body parts hidden from view. Furthermore, performance is limited by the camera's perspective, as detection may fail if key joints are obscured.
- **Limited Suitability for Advanced Users** - The system proved less effective for advanced calisthenics practitioners who may require more complex skills or dynamic feedback.
- **Raspberry Pi 5 Hardware Limitations** - While the Raspberry Pi 5 offers significant performance improvements over earlier models, It is not specifically designed for demanding image processing applications. The device handled pose detection in real time, but exhibited noticeable delays due to the processing load of multiple simultaneous functions and frame analyses.
- **Feedback Restricted to Visual and Audio Cues** - Form correction was limited to on-screen visuals and audio cues. The system did not include detailed voice-guided feedback, haptic or tactile feedback mechanisms.
- **Restricted Operational Distance for Detection** - The system is functioned optimally within a specific calibrated range between the user and the camera. Movements outside this tested range may result in tracking inaccuracies. For the prototype to operate accurately, the user's skeletal frame generated by MediaPipe must remain fully visible within the camera's field of view.

- **No Facial Recognition for Identity Confirmation** - Although user login is facilitated through fingerprint verification, the system does not incorporate facial recognition or any additional biometric authentication methods. This design choice limits the precision of identity verification but ensures user privacy and simplifies the authentication process.
- **Single User Focus** - The device was designed to detect a single user per session. Its detection and analysis features are optimized for individual use, ensuring accurate tracking of posture and form without interference from other individuals in the frame.
- **No Automatic Camera Elevator/Tracker** - The camera elevation system does not feature automatic user tracking; its adjustments are pre-programmed rather than responsive. The linear actuator extends the camera based only on whether the upcoming exercise is designated as "ground based", not in response to the user's real-time position or movement within the frame.

Review of Related Literature

As the global demand for health and wellness continues to grow, individuals are increasingly turning to digital fitness tools to support their physical activity goals. Mobile applications, online workout programs, and wearable technologies have become widespread due to their perceived convenience and affordability. However, the existing body of literature reveals persistent challenges in the effectiveness, engagement, and personalization of these technologies, especially for users with minimal fitness knowledge or limited resources. A significant limitation observed in fitness applications is the absence of real-time corrective feedback. According to Kim and Lee (2023), many fitness apps only provide pre-recorded videos or general instructions that do not adapt to the user's posture or progress. This lack of interactivity creates a risk for beginners who may unknowingly perform exercises incorrectly, leading to inefficient workouts or potential injuries. Garcia and Tan (2023) emphasized that users frequently discontinue using fitness apps due to this lack of support and personalization. Their study revealed that while apps can provide a structured routine, they often fail to assess or correct user performance, resulting in reduced motivation and minimal progress. The danger of incorrect posture during unsupervised exercise has been thoroughly documented. A study by Nordin et al. (2021) noted that improper form, especially in resistance or bodyweight exercises, can lead to musculoskeletal injuries and

chronic pain. This concern is amplified among novice exercisers who may not have prior training in maintaining correct form. In this context, real-time feedback systems offer an effective preventive mechanism. Pose correction systems that use computer vision, such as Pose Trainer developed by Chen and Yang (2020), provide immediate, data-driven feedback by analyzing vector geometry from camera-based pose estimation. These systems are built on tools like OpenCV and OpenPose, which have demonstrated high accuracy in detecting body keypoints and tracking user movement in real time (Cao et al., 2019).

In addition to form-related risks, the high cost of personal trainers and structured gym programs continues to limit access to expert guidance. A market study conducted in Lipa City revealed that fitness coaching sessions can range from ₱650 per session to ₱30,000 annually for gym memberships. While these services offer individualized instruction, they remain financially inaccessible to a significant portion of the population, particularly students and low-income professionals. As a result, many individuals opt for unmonitored home workouts, despite their risks and limitations. To address these challenges, researchers and developers have explored intelligent fitness systems that incorporate artificial intelligence and pose estimation. Zhou et al. (2022) highlighted the effectiveness of AI-powered platforms in improving user form and engagement. Their study suggested that systems capable of real-time feedback can significantly enhance the safety and efficacy of home workouts. Similarly, Yadav et al. (2021) demonstrated how deep learning models, combined with computer vision, could classify user poses and provide adaptive coaching for improved performance. Locally, various studies have explored the barriers and motivations associated with physical activity. Beunaflor et al. (2022), in a study conducted at the University of Batangas, found that academic workload was a major factor contributing to students' limited engagement in physical activity. Araneta et al. (2024) reported that students responded positively to gamified fitness systems that provide structure, guidance, and motivation. These findings indicate a growing interest in fitness technologies that do more than just instruct. They must also engage and personalize the experience for the user. Furthermore, a study by Magnaye et al. (2023) identified physical inactivity as a key factor contributing to obesity among adolescents and adults in Batangas. This highlights the importance of encouraging sustainable and consistent exercise habits through interactive systems that support long-term engagement. In a broader national context, Thim (2023) reported an increase in health awareness among Filipinos, driven by the rising prevalence of lifestyle-related diseases

and greater public access to health information. Other international studies have emphasized the value of real-time coaching systems. For instance, research by Han et al. (2020) proposed the integration of IoT-based posture sensors with smartphone feedback, providing users with correctional data to improve their workout form. Although effective, these systems often rely on costly hardware and are inaccessible to general consumers. Techno-Fitness addresses this gap by utilizing low-cost computer vision systems such as OpenCV and Mediapipe on Raspberry Pi hardware to deliver similar functionality at a more affordable rate. Finally, the motivation to engage in fitness is often linked to internal and external factors. Cagas et al. (2014) noted that appearance, weight control, and physical strength were primary motivators for Filipino youth to engage in exercise. However, without external accountability or support, many individuals find it difficult to maintain consistency in their routines. This reinforces the need for technology-driven systems that can act as virtual coaches by offering adaptive support, feedback, and progress tracking.

To summarize, the literature strongly supports the development of intelligent fitness systems that combine the accessibility of home-based exercise with the precision and safety of guided coaching. The limitations of existing fitness applications, the high cost of personal trainers, and the health risks associated with incorrect form all point to a critical need for systems like Techno-Fitness. By leveraging computer vision through OpenCV and Mediapipe, Techno-Fitness has the potential to bridge these gaps and provide a safe, efficient, and effective workout experience for individuals at the beginner to intermediate level.

Synthesis and Summary

The body of related literature presented a clear picture of the current fitness landscape, particularly in the context of digital solutions and user behavior. While the popularity of fitness applications and online platforms has grown substantially in recent years, especially after the global pandemic, studies consistently highlight significant shortcomings in these tools. Most fitness apps provide only generic workout routines without offering real-time feedback or adjusting to individual user needs. As reported by Garcia and Tan (2023), many users discontinue these apps due to a perceived lack of progress and support, with beginners often finding them inadequate for proper guidance.

International research supports these findings. Kim and Lee (2023) noted that fitness applications are unable to detect if a user is executing an exercise with correct form. This limitation not only reduces workout effectiveness but also increases the risk of injury. The inability to monitor posture is a major flaw, especially in exercises that require precise body alignment. In contrast, real-time pose detection using MediaPipe offers a more dynamic solution. For instance, the Pose Trainer system developed by Chen and Yang (2020) and the OpenPose framework by Cao et al. (2019) demonstrate the effectiveness of computer vision in providing corrective feedback based on pose estimation, using tools such as OpenCV. Local studies further emphasize the challenges faced by the Filipino population in maintaining consistent physical activity. Beunaflor et al. (2022) identified time constraints and academic pressures as major barriers for students. However, studies such as those by Araneta et al. (2024) show a high level of interest in fitness systems that provide support and engagement, especially those with game-based elements. This is particularly important in addressing lifestyle diseases such as obesity, as observed in the study by Magnaye et al. (2023), which links physical inactivity to serious health and psychological consequences in the province of Batangas.

Economic factors also contribute to the problem. Gym memberships and personal training sessions remain financially inaccessible for many, making them unsustainable for long-term fitness. This creates a gap between affordability and the need for professional guidance. The Techno-Fitness device is positioned as a response to this gap by offering a cost-effective, guided system that incorporates real-time pose detection using OpenCV and Mediapipe technology.

In summary, the reviewed literature collectively underscored the urgent need for an accessible fitness solution. Existing mobile applications fall short due to their inability to provide real-time assistance. The high cost of personal trainers and gyms creates additional barriers. Meanwhile, technological advancements in pose estimation provide a viable alternative. The Techno-Fitness project directly addresses these issues by combining affordability with a live workout interface catering audio and visual cues, making it particularly suitable for beginner to intermediate users who prefer home-based calisthenics workouts. This integration of technology not only improves workout safety and effectiveness but also promotes long-term engagement in physical activity.

Theoretical Framework

The Techno-Fitness device is grounded in multiple theories that support its development and effectiveness in promoting fitness, motivation, and user engagement. This framework integrates principles from behavior change theories, human-computer interaction, and artificial intelligence to justify the research's approach to guiding users in their fitness journey. With this, one of the primary theories relevant to this research is ***Bandura's Social Cognitive Theory***, wherein it emphasized the role of observational learning, self-efficacy, and feedback on behavior change. The Self-Determination Theory claims that autonomy, competence & relatedness are fundamental psychological needs driving motivation & personal growth as fostering environments that promote and support these needs can enhance one's motivation, leading to improved performance and well-being (Ackerman, 2018). In relation to the research, this emphasized the role of observational learning, self-efficacy, and feedback in behavior change as the device provides real-time feedback and assistance, reinforcing correct exercise habits and increasing efficiency in their fitness routines. By incorporating Mediapipe that detects posture, the device helps users develop self-regulated fitness behaviors. Another theoretical framework that the research would touch up is the ***Fogg's Behavior Model*** wherein it is a universal way of thinking about the drivers of behavior change, identifying three critical elements that must converge simultaneously for a behavior to occur: *Motivation, Ability, and Prompt* (Lau, 2025). The research shall delve into the three key factors that support the Fogg Behavior Model. In the first factor, motivation is where the users are to stay consistent with their fitness routines by receiving real-time assistance, progress tracking, and structured workout plans through the usage of the device. The second factor, ability, refers to the ease of access and progressive workouts provided to the users. By removing barriers such as lack of knowledge or uncertainty about proper form, the system increases the likelihood of users sticking to their fitness goals. Lastly, prompt serves as the trigger that encourages users to engage in their workouts consistently. The system achieves this by implementing engaging features such as letting users see the pose detection, audio and visual cues, and a grading system that would prompt users to perform better in their sessions. This ensures that individuals remain consistent and develop a structured routine over time. Since an individual would be highly involved into the usage of the device, it is related to another of Fogg's theories which is the ***Persuasive Technology Theory*** shall come into play. This theory is any digital system that influences user behavior or attitudes (Arora, 2024). By



applying the Persuasive Technology Theory, the research aims to guide users through a comprehensive fitness journey, utilizing motivational elements, simplifying tasks, and incorporating a grading system that encourages regular exercise. This is to effectively reduce barriers to entry, sustain user engagement, and create long-term behavioral changes that promote a healthier lifestyle.

MATERIALS AND METHODS

Methods of Research

This section outlines the approach undertaken in developing and evaluating the Techno-Fitness device, detailing the processes of development, participant involvement, research design, data collection methods, and data analysis procedures that ensured the validity and relevance of the results. The development of the device followed a systematic process consisting of seven stages: *Planning Stage, Data Collection, Conceptualization, Initial Prototyping, Testing and Evaluation, Refinement, and Final Implementation*. The study employed a Developmental Research approach, which focused on the systematic design, development, and evaluation of a technology-based solution specifically, the Techno-Fitness Device.

The project began with the planning phase, which established a solid foundation for the development of the device. During this phase, the research team identified the primary problems, reviewed related literature, and formulated objectives that guided the entire development process. Following this, the Data Collection phase was conducted. This phase implemented the developmental research design wherein both qualitative and quantitative data were obtained. The qualitative data were gathered through a consultation with a fitness trainer, while the quantitative data were collected from survey responses. This mixed-methods design provided comprehensive insights into both the device's technical functionality and user experience, contributing significantly to its final refinement and successful implementation.

Data Gathering

Moving onto qualitative data gathering, a consultation period with a fitness coach served as the primary source of qualitative data to validate the system's relevance, technical direction, and practicality. A semi-structured interview format was utilized to allow in-depth exploration of the coach's professional insights, while maintaining consistency across responses. The qualitative consultation with **Engr. Michael Guantero**, a licensed engineer and certified fitness coach with years of experience in designing training programs for various fitness levels, He assessed the selected exercises to ensure their suitability for beginner to intermediate users and verified that they aligned with standard safety and effectiveness guidelines. His expertise helped in defining correct posture and movement criteria, which guided the system's real-time assistance feature. He also suggested potential feature enhancements such as nutritional infographics, and warm-up/cool-down modules to improve the system's functionality. This collaboration ensured that the developed device not only functions on a technical level but also upholds professional standards expected in real-world fitness coaching. The insights gathered from this consultation were organized and analyzed to identify key findings.

One of the most significant takeaways from the interview was the emphasis on the balance between nutrition and physical activity. According to the fitness experts, achieving one's fitness goals is approximately 70% dependent on diet and 30% on exercise. This finding highlighted that proper nutrition plays a crucial role in transforming and maintaining the body, as dietary habits significantly influence energy levels, recovery, muscle development, and overall health. Therefore, while exercise is essential for strength, endurance, and cardiovascular health, it is ultimately the food intake that determines how effectively the body responds and adapts to physical training as one saying says "You are what you eat". In response to this, the researchers integrated a supplementary feature in the system that displays nutritional infographics to educate and remind users about proper diet and health management. Additionally users would be able to track their history and monitor their progress.

The fitness trainer also pointed out that this device would be more inclined into developing the fitness of beginners compared to people whose fitness level are of that advanced level. The said people will have the autonomy, knowledge and confidence to know the correct postures are and what exercise to do. As the fitness coach mentioned that if a person's fitness



level is higher than those who are beginners, it would be disadvantageous for them as it would be more efficient if they were to formulate their own workout plans that are tailored to their goals. However, he also said that it would be very beneficial for beginners as well as other fitness enthusiasts to utilize this device as it can be a stepping stone for their fitness journey as it would be able to teach them the basics and the fundamentals of fitness. Based on his experience, he shared that when he asked one of his clients to perform familiar exercises such as squats and push-ups, exercises the client claimed to know well, the execution revealed improper form. Despite the client's confidence, key elements of posture and technique were incorrect, which could lead to ineffective results or even injury. This underscores a common issue among individuals who exercise without professional guidance: while they may be aware of certain exercises, they often lack the proper form needed to perform them safely and effectively. As such, this reinforces the importance of targeting beginners as the primary users of the Techno-Fitness System.

Table 1.1
Calisthenics Week 1 Workout Plan

Week 1				
Day	Exercises	Reps/Time	Sets	Rest
Day 1	Knee Push Ups	8 reps	3	20s
	Body Weight Squats	12 reps	3	20s
	Plank (engage core, flat back)	30 seconds	3	20s
	Glute Bridges	12 reps	3	20s
	Bent Over Rows	8 reps	3	20s
	Leg Raises	10 reps	3	20s
	Superman Hold	20 seconds	3	20s
Day 2	Glute Bridges	15 reps	2	45s
	Leg Raises	12 reps	2	45s
	Body Weight Squats	15 reps	2	45s
	Plank	30 seconds	2	45s
	Superman Hold	20 seconds	2	45s
Day 3	Knee Push Ups	10 reps	2	60s
	Bent-over Rows (Bodyweight)	10 reps	2	60s
	Plank	30 seconds	2	60s



	Superman Hold	20 seconds	2	60s
	Knee Push Ups	8 reps	2	60s
Day 4	Knee Push Ups	6 reps	3	30s
	Body Weight Squats	10 reps	3	30s
	Plank (don't let hips sag)	20 seconds	3	30s
	Glute Bridges	10 reps	3	30s
	Bent-over Rows (Bodyweight)	6 reps	3	30s
	Leg Raises	12 reps	3	30s
	Superman Hold	15 seconds	3	30s
Day 5	Knee Push Ups	12 reps	2	45s
	Body Weight Squats	15 reps	2	45s
	Leg Raises	12 reps	2	45s
	Glute Bridges	15 reps	2	45s
	Bent-over Rows (Bodyweight)	12 reps	2	45s
	Leg Raises	12 reps	2	45s
	Superman Hold	20 seconds	2	45s
Day 6	Leg Raises	12 reps	2	45s
	Plank	30 seconds	2	45s
	Bent Over Rows	10 reps	2	45s
	Glute Bridges	12 reps	2	45s
	Superman Hold	20 seconds	2	45s
Day 7	Rest Day	-	-	-

Table 1.2
Calisthenics Week 2 Workout Plan

Week 2				
Day	Exercises	Reps/Time	Sets	Rest
Day 1	Knee Push Ups	10 reps	2	45s
	Body Weight Squats	14 reps	2	45s
	Plank	35 seconds	2	45s
	Glute Bridges	14 reps	2	45s
	Bent-over Rows (Bodyweight)	10 reps	2	45s
	Leg Raises	12 reps	2	45s
	Superman Hold	25 seconds	2	45s
Day 2	Glute Bridges	17 reps	2	45s



	Leg Raises	14 reps	2	45s
	Body Weight Squats	17 reps	2	45s
	Plank	35 seconds	2	45s
	Superman Hold	25 seconds	2	45s
Day 3	Knee Push Ups	12 reps	3	60s
	Bent-over Rows (Bodyweight)	12 reps	3	60s
	Plank	35 seconds	3	60s
	Superman Hold	25 seconds	3	60s
	Knee Push Ups	10 reps	3	60s
Day 4	Knee Push Ups	8 reps	1	30s
	Body Weight Squats	12 reps	1	30s
	Plank	25 seconds	1	30s
	Glute Bridges	12 reps	1	30s
	Bent-over Rows (Bodyweight)	8 reps	1	30s
	Superman Hold	20 seconds	1	30s
Day 5	Knee Push Ups	14 reps	3	30s
	Body Weight Squats	17 reps	3	30s
	Glute Bridges	17 reps	3	30s
	Plank	35 seconds	3	30s
	Bent-over Rows (Bodyweight)	14 reps	3	30s
	Leg Raises	14 reps	3	30s
	Superman Hold	25 seconds	3	30s
Day 6	Leg Raises	14 reps	2	45s
	Plank	35 seconds	2	45s
	Bent-over Rows (Bodyweight)	12 reps	2	45s
	Glute Bridges	14 reps	2	45s
	Superman Hold	25 seconds	2	45s
Day 7	Rest Day	-	-	-

Table 1.3
Calisthenics Week 3 Workout Plan

Week 3				
Day	Exercises	Reps/Time	Sets	Rest
Day 1	Push Ups	12 reps	3	60s



	Body Weight Squats	15 reps	3	60s
	Plank	40 seconds	3	60s
	Glute Bridges	15 reps	3	60s
	Bent-over Rows (Bodyweight)	12 reps	3	60s
	Leg Raises	12 reps	3	60s
	Superman Hold	30 seconds	3	60s
Day 2	Glute Bridges	20 reps	3	45s
	Leg Raises	15 reps	3	45s
	Body Weight Squats	20 reps	3	45s
	Plank	40 seconds	3	45s
	Superman Hold	30 seconds	3	45s
Day 3	Push Ups	14 reps	3	60s
	Bent-over Rows (Bodyweight)	14 reps	3	60s
	Plank	40 seconds	3	60s
	Superman Hold	30 seconds	3	60s
	Push Ups	12 reps	3	60s
Day 4	Push Ups	10 reps	3	30s
	Body Weight Squats	15 reps	3	30s
	Plank	30 seconds	3	30s
	Glute Bridges	15 reps	3	30s
	Bent-over Rows (Bodyweight)	10 reps	3	30s
	Leg Raises	15 reps	3	30s
	Superman Hold	25 seconds	3	30s
Day 5	Push Ups	15 reps	3	30s
	Body Weight Squats	20 reps	3	30s
	Glute Bridges	20 reps	3	30s
	Plank	40 seconds	3	30s
	Bent-over Rows (Bodyweight)	15 reps	3	30s
	Leg Raises	15 reps	3	30s
	Superman Hold	30 seconds	3	30s
Day 6	Leg Raises	15 reps	2	45s
	Plank	40 seconds	2	45s
	Bent Over Rows (Bodyweight)	14 reps	2	45s
	Glute Bridges	15 reps	2	45s
	Superman Hold	30 seconds	2	45s



Day 7	Rest Day	-	-	-
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Table 1.4
Calisthenics Week 4 Workout Plan

Week 4				
Day	Exercises	Reps/Time	Sets	Rest
Day 1	Push Ups	15 reps	3	60s
	Body Weight Squats	20 reps	3	60s
	Plank	45 seconds	3	60s
	Glute Bridges	20 reps	3	60s
	Bent-over Rows (Bodyweight)	15 reps	3	60s
	Leg Raises	15 reps	3	60s
	Superman Hold	35 seconds	3	60s
Day 2	Glute Bridges	22 reps	3	45s
	Leg Raises	17 reps	3	45s
	Body Weight Squats	22 reps	3	45s
	Plank	45 seconds	3	45s
	Superman Hold	35 seconds	3	45s
Day 3	Push Ups	17 reps	3	60s
	Bent-over Rows (Bodyweight)	17 reps	3	60s
	Plank	45 seconds	3	60s
	Superman Hold	35 seconds	3	60s
	Push Ups	15 reps	3	60s
Day 4	Push Ups	12 reps	3	30s
	Body Weight Squats	17 reps	3	30s
	Plank	35 seconds	3	30s
	Glute Bridges	17 reps	3	30s
	Bent-over Rows (Bodyweight)	12 reps	3	30s
	Leg Raises	17 reps	3	30s
	Superman Hold	30 seconds	3	30s
Day 5	Push Ups	20 reps	3	30s
	Body Weight Squats	22 reps	3	30s
	Glute Bridges	22 reps	3	30s
	Plank	45 seconds	3	30s
	Bent-over Rows (Bodyweight)	17 reps	3	30s



	Leg Raises	17 reps	3	30s
	Superman Hold	35 seconds	3	30s
Day 6	Leg Raises	17 reps	3	45s
	Plank	45 seconds	3	45s
	Bent-over Rows (Bodyweight)	15 reps	3	45s
	Glute Bridges	17 reps	3	45s
	Superman Hold	35 seconds	3	45s
Day 7	Rest Day	-	-	-

The figures above outline a 4-week progressive calisthenics program that is beginner-friendly, using only bodyweight exercises to build strength, improve mobility, and enhance core stability. The plan follows a structured 7-day cycle (6 workout days and 1 rest day) that advances each week. It features a core set of exercises (like push-ups, squats, planks, and glute bridges) that are combined differently each day to provide a varied focus, targeting the full body, upper body, lower body, and core. No special equipment is required, making it ideal for home workouts. The program ensures safe progression by gradually increasing reps, durations, and exercise difficulty. For example, it starts with Knee Push Ups for the first two weeks and advances to standard Push Ups for weeks 3 and 4. Each workout includes clear sets, reps, and rest intervals, with lighter-intensity days also built in to promote consistency and proper recovery.

Table 2
Warm-up Exercises & Cooldown Exercises

Warm-Ups	
Exercise	Reps/Time
Arm Circles	20 seconds
Jumping Jacks	12 reps
Cross Body Stretches (Left)	30 seconds
Cross Body Stretches (Right)	30 seconds
Hip Circles	20 seconds
Body Weight Lunges (Left)	10 reps
Body Weight Lunges (Right)	10 reps
Shoulder Rolls	20 seconds



Cooldowns	
Exercise	Time
Standing Hamstring Stretch	30 seconds
Standing Leg Stretches (Left)	30 seconds
Standing Leg Stretches (Right)	30 seconds
Cobra Stretch	30 seconds
Quad Stretch (Left)	30 seconds
Quad Stretch (Right)	30 seconds

To maximize the effectiveness of any workout, it's essential to prepare the body properly and recover afterward, and this routine provides a clear plan for both. The Warm-Up is designed to increase heart rate and prepare the body for movement using a combination of dynamic exercises (like Arm Circles and Hip Circles), rep-based exercises (like Jumping Jacks and Body Weight Lunges), and static stretches (like Cross Body Stretches). This mix activates key muscle groups and loosens joints to reduce injury risk and boost performance. Following the workout, the Cooldown helps the body recover by focusing purely on static stretches such as the Standing Hamstring Stretch, Cobra Stretch, and Quad Stretches, to help lower the heart rate, relieve muscle tension, and promote flexibility.

Hardware & Software Design Development

Software Design

Figure 1.1
Login and Main Menu Interface

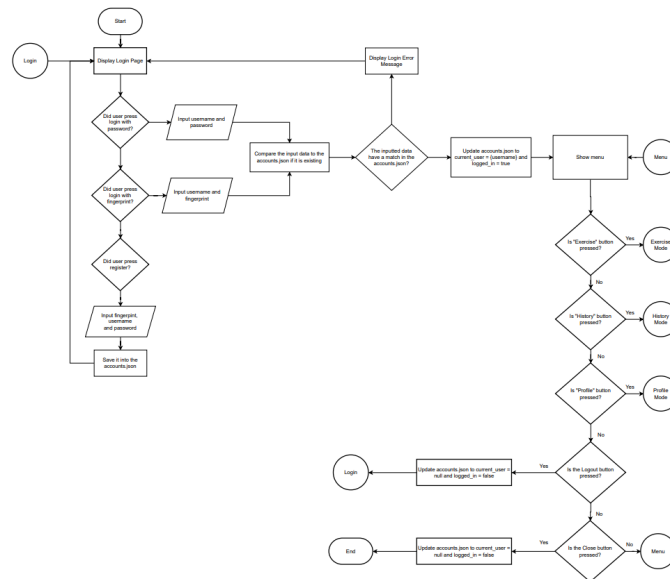


Figure 1.2
Workout and Prerequisites Menu Interface

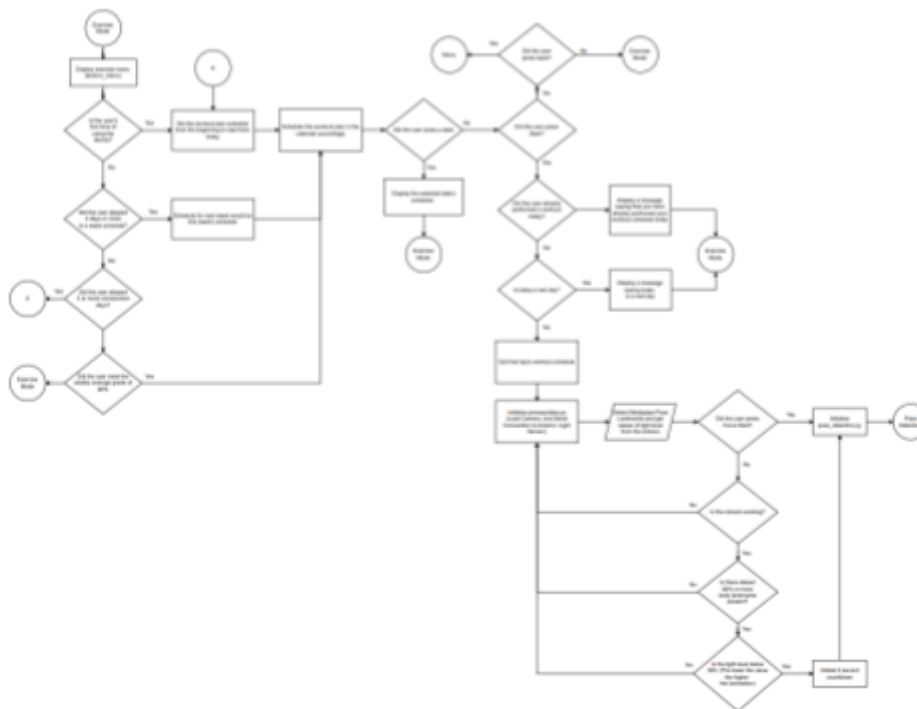
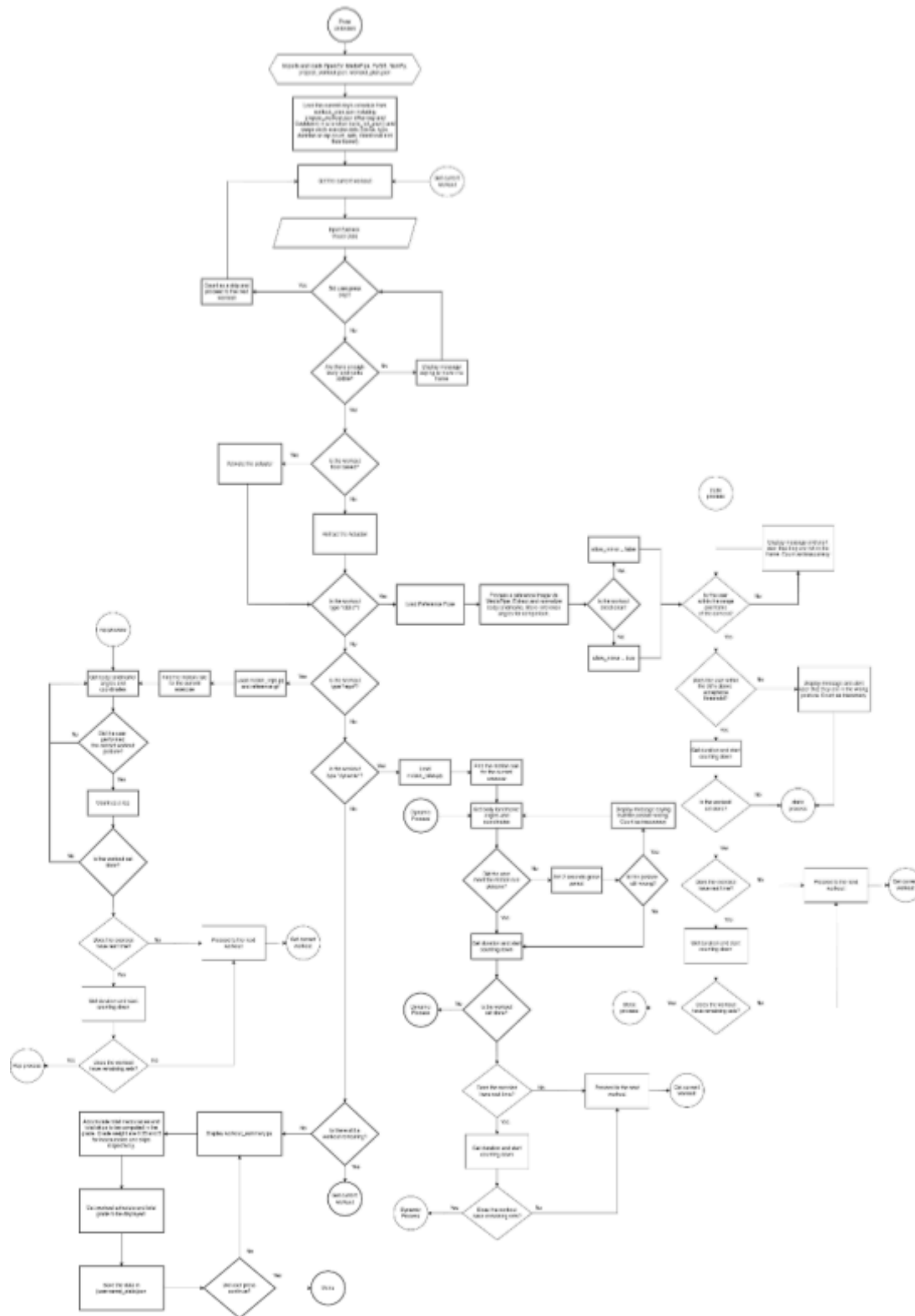


Figure 1.3
Posture Detection and Workout Summary Interface



This phase marked the transition from planning to the completion of the Techno-Fitness prototype. To bring the concept into reality, a comprehensive range of hardware and software components was utilized. The completed Techno-Fitness device has resulted in a functional prototype that is capable of detecting postures or forms of different exercises and tracking performance in real time. The researchers implemented the system using the Raspberry Pi 5, which required the installation of several essential libraries such as PyQt6, heavily used in developing the user interface. The integration of OpenCV enabled the MediaPipe framework to accurately detect the body landmarks through a live camera feed. Since computers cannot directly interpret visual images, OpenCV served as the intermediary, translating visual information into data that computers can read. In this system, OpenCV functions as the "bridge," capturing raw video frames from the camera, performing necessary preprocessing such as color space conversion (BGR to RGB) and resizing, then displaying the results. This process ensures that the visual data are properly prepared for analysis. Mediapipe, on the other hand, then takes these prepared frames and acts as the "analyzer," employing its pose estimation model to identify and locate key body landmarks (joints) within the image. The extracted body landmark data served as the core output used for calculating angles and comparing the user's posture against defined rules or reference images. Together, OpenCV manages the image input and output pipeline, while Mediapipe extracts and interprets the crucial human pose information from those images. The device determines whether a certain posture is correct by assigning an exercise type for each workout. There are three types of exercise applied in the program which are *static*, *reps*, and *dynamic*. These types enable the program to be able to easily distinguish between what rules would the program apply for the detection of the correct posture of an exercise.



Figure 2.1: Installation of Libraries in Raspberry Pi 5

1. *Static exercises* are exercises that require the user to follow a reference picture accurately with a slight acceptable threshold. Out of the three types, this is a straight forward detection of a correct posture as the user doesn't need to move as they would be doing

one pose for a certain amount of time. For every static exercise, there is only one function that handles the function of correct pose detection as it only needs one reference picture. This one reference picture would have to go through the OpenCV first then onto the Mediapipe wherein it would detect the posture of the reference picture, and extract its angular data to be compared to the current posture that the user is doing through the live feed. If the exercise being done is a directional exercise (e.g. arm stretches, lunges), it would strictly get the correct angular data in respect to the direction being shown in the reference image meaning that the user must exactly do what is being shown in the image. However, if the exercise is non-directional (e.g. planking), it would detect the correct posture regardless of the direction. When being compared, there is a match percentage the user must meet that being 93% and above. If it is below the acceptable threshold, the timer would stop and the program would mark it as inaccurate.

2. *Repetitive and dynamic exercises* are characterized by continuous movement, requiring constant monitoring of joint angles and limb coordination. Unlike static exercises, which maintain a fixed posture and can rely on a single reference image or pose, dynamic and repetition-based movements demand individualized detection logic. Each workout must have its own coded posture evaluation because the indicators vary widely across movements. For example, push-ups emphasize elbow flexion-extension and torso alignment, squats depend on the hip-knee angular relationship and vertical displacement, while arm circles focus on the continuous rotation and range of motion of the shoulders and arms. These variations mean that no universal detection rule can apply to all exercises. Instead, each must be assigned specific motion parameters such as angle thresholds, displacement ranges, or timing patterns. This customized implementation ensures that the system is able to distinguish correct form from incorrect execution, reinforcing the precision of real-time feedback and preventing false detections during diverse movements.

With this, the posture detection through Mediapipe which leverages OpenCV was implemented and was able to detect workout posture of the user. Particularly with the application of the static exercises. It compares the reference image to the live feed of the camera, compares both angular data, and outputs whether the posture is within the acceptable threshold. The

researchers first applied the implementation of the static exercises to gauge and test the capabilities of the Mediapipe framework. In the initial implementation of the static exercise, there was no directional detection in certain exercises meaning that as long as the user does the pose accurately, the match percentage would meet the ninety-three percent (93%) threshold. Before the current accepted threshold, the researchers



Figure 2.2: Testing of the Initial Prototype

conducted multiple calibration tests using varying similarity percentages to determine the most suitable threshold for posture accuracy. Initially, lower thresholds varying from 60% to 90% were tested, but these values resulted in frequent false positives instances where incorrect postures were mistakenly classified as correct. Conversely, higher thresholds such as 95% and above were too restrictive, causing the system to reject postures that were visually acceptable. Through iterative testing it was determined that a ninety-three percent (93%) threshold provided the best balance between precision and tolerance. In addition, it provides the optimal balance between accuracy and flexibility in posture recognition. This threshold allows slight variations in body alignment without penalizing users for minor deviations caused by differences in physique or camera angle. This is to also cater to the detection limits of the Mediapipe framework where minor fluctuations in landmark positioning can occur due to factors such as lighting variations, camera resolution, and partial occlusion of body parts. These slight inconsistencies can affect the precision of angle calculations and, consequently, the similarity score. The detection of the static exercises paved the way for the other exercise types. These slight inconsistencies can affect the precision of angle calculations and, consequently, the computed match score. Despite these challenges, the successful detection and evaluation of static exercises served as a foundational milestone for the system's development. This phase not only validated the accuracy and responsiveness of Mediapipe in recognizing human posture but also established the baseline for implementing more complex exercise types such as dynamic and repetition-based movements. Through this initial implementation, the researchers effectively achieved Objective Number One (1), to explore and test the posture detection capabilities of the Mediapipe framework. By

starting with simple static exercises such as stretching, the process revealed the framework's strengths and limitations, allowing for the refinement of angle detection algorithms and confirming Mediapipe's viability as the core component for real-time posture monitoring within the Techno-Fitness system.

Moving on, repetition-based exercises were identified as the next most straightforward exercise type to develop following static exercises, primarily because such workouts typically involve only two distinct movement states. For example, jumping jacks operate under a motion rule composed of two decision logics used for rep counting. When both arms are raised and slightly bent while the legs are spread apart, the system recognizes this as half a repetition. Conversely, when the arms return to the sides and the legs close together, the system records a complete repetition. This logic ensures that only movements matching the predefined motion rule are counted, preventing unrelated gestures or incorrect postures from being falsely recognized as valid reps. However, a minor challenge arises when a user's movement closely resembles the defined motion rule, as the system may occasionally register a false positive. For instance, with the case of push ups. Its motion involves a lowering phase where the elbows bend which counts as a half a rep, bringing the chest towards the floor, followed by an upward phase where the arms extend to push the body back to the starting plank position which counts as a full rep. A false positive might occur if the user performs a movement that partially mimics this sequence without completing a full push-up. This is especially true if the motion rule relies heavily on just one joint, like the elbow bending and extending such as curls. If the motion rule for push-ups isn't specific enough about the range of elbow/shoulder angles required for both the down and up phases, or the sequence in which they must occur, these similar but incorrect movements could accidentally trigger a rep count.

With this, it was also observed that Mediapipe's posture detection can exhibit inconsistencies, particularly when certain body joints are not clearly visible to the camera. Missing landmark data can lead to incomplete angular calculations, reducing the accuracy of pose evaluation. The findings revealed that while Mediapipe performs reliably in well-lit and unobstructed



Figure 2.3: Testing of Dynamic Exercises

environments, its accuracy may decrease when visibility issues occur, and when another person is visible within the frame. This emphasizes the importance of proper positioning, lighting, and camera calibration in achieving consistent detection results.

Dynamic exercises in Techno-Fitness were successfully recognized through continuous tracking and analysis of body landmarks, emphasizing on how these points move across consecutive frames to form specific motion patterns. Unlike static or repetition-based exercises, that relied on posture matching or discrete movement cycles, dynamic exercises involve uninterrupted motion that must be identified and validated in real time. The system continuously monitors the user's joint trajectories, angular changes, and movement direction to determine whether the motion aligns with the predefined pattern of the targeted exercise. For instance, in the case of arm circles, the detection process functioned differently from that of static or rep-based movements. Instead of an “up–down–up” repetition pattern, the system evaluates the smooth, rotational continuity of the arm's movement. Using Mediapipe integrated with OpenCV, the system extracts the coordinates of key landmarks such as the shoulder, elbow, and wrist. It then computes the wrist's angular displacement relative to the shoulder with respect to the horizontal axis. As the user performs circular rotations, the computed angle changes dynamically between frames. The program continuously compares the current angle to the previous one, determining whether the rate and direction of change remain within the expected range that signifies a circular motion. When the system detects consistent angular displacement that aligns with the expected circular path and speed, it classifies the action as a valid arm circle motion, confirming that the user is performing the exercise correctly. This process repeats throughout the activity, providing real-time tracking and feedback to ensure the user maintains continuous movement. It is important to note, however, that this motion rule is unique to arm circles. A two-second grace period mechanism was also implemented to enhance detection stability during dynamic workouts. This feature prevented abrupt fluctuations in detection status by allowing the system to verify any posture change before updating the feedback. Without this buffer, brief inconsistencies such as camera misreads, MediaPipe glitches, or transient joint tracking delays, could have caused the system to rapidly alternate between “correct” and “incorrect” states, leading to unreliable feedback. During this two-second grace period, the system continues to analyze the movement frame by frame using OpenCV and Mediapipe landmark data. If the

motion remains stable and within the expected threshold throughout the grace period, the system confirms the posture and proceeds to progress the duration timer. However, if the movement remained wrong even after the grace period, it would be detected as an inaccuracy/wrong and the timer would stop. It is similar to the duration timer of the static exercises wherein if the user were to perform the correct exercise posture, the duration timer would progress again.

With this implementation, each exercise was integrated into the Raspberry Pi 5 using JSON files as the primary data structure for account data saving, storing workout parameters, and reference postures. which focuses on applying posture detection feedback through the use of the Mediapipe framework for each exercise in the workout plan. By structuring the exercise data in a JSON format, the system was able to efficiently retrieve the motion rule parameter and address of the reference image which allows the system to process posture references in real time, allowing seamless communication between the pose detection module and the exercise logic. This design choice was made to achieve optimal system performance by minimizing computational overhead. JSON files provide lightweight and direct data access, allowing the pose detection module to efficiently retrieve exercise information without the latency associated with database queries. Utilizing a traditional database would have imposed additional processing demands on the Raspberry Pi 5, which operates within limited hardware resources compared to full-scale computing systems. During testing, it was determined that monitoring frames and real-time responsiveness in posture detection required allocating as much of the device's processing power as possible to the Mediapipe and OpenCV pipelines. By avoiding continuous database interactions, the system ensures smoother frame analysis, reduced latency, and more stable feedback generation.



```
{
  "prepost": "Cooldown",
  "exercises": [
    {
      "name": "Deep Breathing",
      "type": "dynamic",
      "duration": 60,
      "motion_rule": "deep_breathing",
      "rest_time": 0,
      "floor_based": false,
      "ref_image": "resources/pose/ref_armcircles.jpg"
    },
    {
      "name": "Standing Toe Touch Stretch",
      "type": "static",
      "duration": 30,
      "ref_image": "resources/pose/ref_standingtoetouch.jpg",
      "rest_time": 10,
      "directional": false,
      "floor_based": false
    }
  ]
}
```

Figure 2.4: Snippet of Cooldown Exercises

The prototype leverages JSON files for persistent data storage, managing critical information such as user login status and session details, the structured list of workouts organized by week and day, and individual user performance history and progress. This user

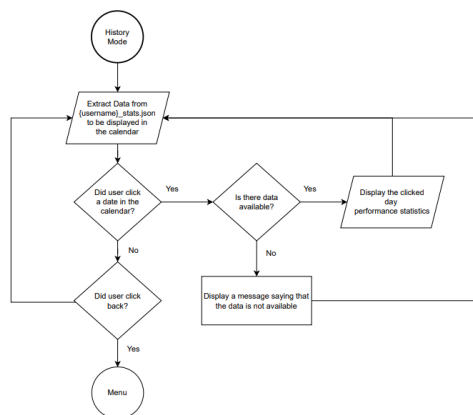
statistics file notably stores the crucial start date for their calisthenics journey (defaulting to the first day they would use the program if not set), their last workout date, and potentially weekly stats including grades. The workout menu interface features a prominent calendar system which visually maps out the user's workout schedule. This isn't just a static display; the schedule dynamically adjusts based on the user's adherence and performance, reflecting the concept of effective workout progression rather than just calendar time. Users can click on any date to view the planned workout details for that day in a dialog. A core feature implemented in this menu is the adaptive scheduling logic, primarily handled by the imported scheduler module and methods like *calculate_skip_info* and *compute_effective_week_and_day_indices*. The system enforces conditions to ensure consistency:

- If a user skips five (5) or more consecutive days (determined by comparing today's date to the last workout date in their stats), their schedule progress is reset, effectively setting their *start_date* to the current day, making them begin from Week 1, Day 1 again, although their past history remains.
- If a user skips three (3) or more workout days within a completed week, or if their calculated weekly grade (potentially read from *weekly_stats*) falls below a passing threshold (e.g., 60%), that specific week's schedule is marked for repetition.
- This repeat mechanism intelligently shifts subsequent weeks; for instance, if week one (1) is flagged for repetition, the workout plan originally intended for the calendar's week two (2) will be effectively pushed to the user's third calendar week, ensuring mastery before progression. The UI provides feedback on this status above the start button.

When the user initiates a workout via the "Start" button, the system first checks if the current day is designated as a rest day or if the workout for the day has already been completed. If neither is true, it calculates the correct effective week and day number for the current date, considering their start date and any necessary repetitions. It then launches the prerequisites menu screen as a separate process, passing the calculated week, day, and the current username. The prerequisites screen handles camera, lighting level, and visible landmark visibility checks before subsequently launching the main pose detection workout interface.

Consequently, after a workout session, the user is then redirected to the workout summary page. Its primary function is to calculate and persist performance metrics for that session. The workout summary lists the exercises performed, the number of exercises skipped, and count of total inaccuracies and wrongs. Upon initialization, it calculates a grade for the completed session using predefined weights for inaccuracies and skips. The workout grade starts at a maximum of 100 points. Points are then deducted based on performance issues during the session: 0.35 points are subtracted for each detected inaccuracy, and 5 points are subtracted for each exercise skipped. The final daily average grade is calculated using the formula: $100.0 - (\text{Total Wrongs} * 0.35 + \text{Total Skips} * 5.0)$. If this calculation results in a score below zero, the grade is floored at 0.0. This daily grade is then saved, and it contributes to the calculation of the week grade (the average of grades within that week) and the overall grade (the average across all recorded days). Crucially, it then loads the current user's statistics file, updates the entry for the current day with the session's date, total wrongs, total skips, and the calculated average_grade. It also recalculates and updates the corresponding week grade and the overall overall grade within the statistics file before saving it back to disk. The UI displays the list of exercises performed in the session, the calculated statistics for the day (grade, wrongs, skips), and provides qualitative remarks based on the achieved grade.

Figure 3.1
History Interface



Moving on, the figure above shows the history tab. It is a module enabling users to review their past performance and track progress. The page heavily includes a calendar widget to which can be interacted with by the user. Beside it is a details panel containing information about their performance. Clicking a calendar date populates the right panel with details for that day, including the average grade, total wrongs, total skips, overall grade, and calculated percentage

improvement compared to the previous day. The page also features a 7-day bar chart of recent daily grades. With this, it gets the logged in user's JSON data statistics to be displayed in the page. It then visually marks dates with workout data on the calendar. The bar graph visualizes the last seven days' grades, color-coding bars based on performance and indicating rest/skipped days.

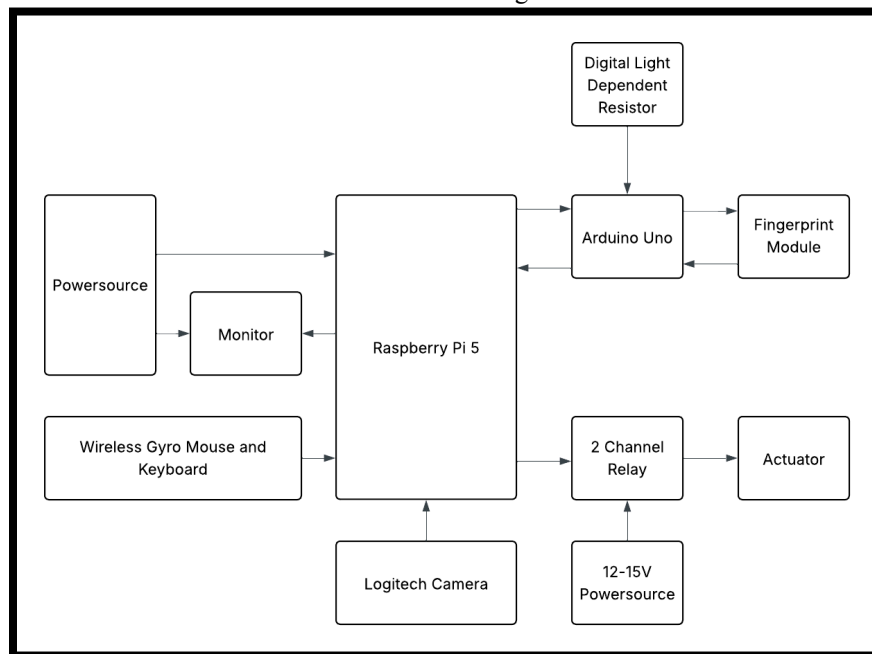
Figure 3.2
Profile Interface



The figure above shows the process flow for the profile module. It defines the User Profile screen, centered on account management and displaying summary progress. Its essential functions allow the user to view their current effective week/day in the workout plan (accounting for repeats). Crucially, it provides options to reset workout progress (by overwriting the user's stats JSON file with defaults) or delete the entire account. The account deletion process includes removing the user from accounts.json, deleting their stats file, and sending a serial command to an Arduino to erase the associated fingerprint data. This screen also acts as a gateway to the Information Portal).

Hardware Design Development

Figure 4
Device Block Diagram

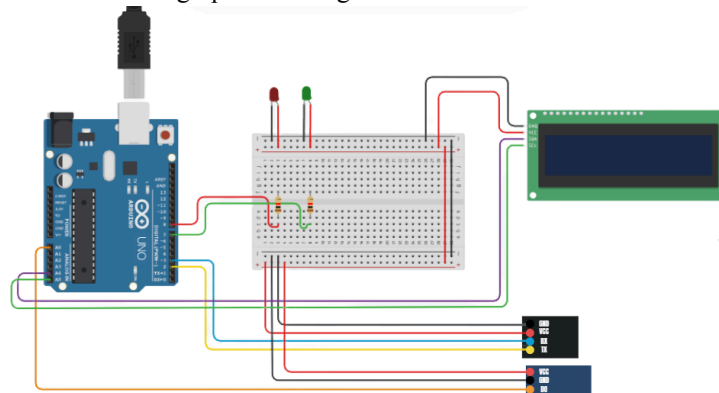


The schematic diagram illustrates the overall system architecture of the Techno - Fitness Prototype, the diagram shows how each component interacts to deliver posture detection, exercise tracking, and feedback functionalities. At the center of the system is the Raspberry Pi 5, which serves as the main processing unit. The Raspberry Pi 5 coordinates data flow between hardware components and executes the posture recognition algorithms developed using OpenCV and Mediapipe. A Logitech Camera is connected to the Raspberry Pi 5 to capture the user's movements in real time. The captured video frames are processed by the Raspberry Pi 5 to detect body posture and form during exercises through the openCV library. More so, a monitor is also connected to the Raspberry Pi 5, in which it will display the interface and visual feedback for the user, while the wireless gyro mouse and keyboard allow convenient system navigation and control. Both the Raspberry Pi 5 and Monitor are powered by the main power source. The Arduino Uno acts as a microcontroller interface responsible for handling peripheral sensors and modules. It receives input from the digital - light dependent resistor to detect lighting conditions and from the fingerprint module, which verifies users identification before system activation. The Arduino Uno communicates bi-directionally with the Raspberry Pi 5 through serial communication in order to synchronize sensor data handling and control commands. A 2-channel

relay module is connected to the Raspberry Pi to control the actuator, which provides mechanical movement (such as adjusting the elevation of the camera).

The hardware functionalities of the *Techno-Fitness* prototype focus on three major aspects. The first involves the account management system, which includes the processes of account creation, verification or login, and deletion. This function allows users to securely register and access their personalized workout data within the system. It ensures that each user's information, such as performance records and fitness progress, is stored and retrieved efficiently. The second functionality pertains to camera elevation, which enables the adjustment of the camera's position to properly capture the user's body posture and movements during exercise. This feature ensures that the system obtains accurate visual input for motion tracking and posture analysis for both ground-based and standing-based exercises. The third functionality focuses on light detection, which is used to monitor environmental brightness during exercise sessions. This feature allows the system to automatically detect insufficient lighting conditions that could affect camera visibility and motion tracking accuracy. By identifying changes in light intensity, the prototype can prompt the user to adjust the lighting setup, ensuring that the posture detection module continues to perform effectively under various illumination conditions.

Figure 5
Fingerprint and Light Detection Circuit



For the account functionality, an R305 Fingerprint Sensor is utilized to capture the user's fingerprint template. The sensor's receiver (RX) and transmitter (TX) pins are connected to pins 2 and 3 of the Arduino board, allowing serial data transmission between the two components. The data collected by the sensor is processed by the Arduino microcontroller, which then

communicates with the Raspberry Pi 5 through serial communication. This configuration enables efficient and secure transmission of authentication data between the two devices, ensuring accurate identification of registered users. To provide visual feedback during the authentication process, two LED indicators and a 16x2 LCD display are integrated into the system. The green LED, connected to pin 7 of the Arduino, lights up when authentication is successful, while the red LED, connected to pin 8, illuminates when an invalid fingerprint or failed verification occurs. The LCD module, which is connected through the I2C interface using pins A4 (SDA) and A5 (SCL), displays corresponding system messages indicating whether the fingerprint verification was successful or unsuccessful, providing the user with clear real-time feedback during the login process.

The process of fingerprint authentication follows a structured flow similar to a flowchart. It begins when the user places their finger on the fingerprint sensor. The R305 module scans the fingerprint and converts the image into a digital template. The Arduino receives this data and compares it with the stored templates in its memory. If the fingerprint matches an existing template, the Arduino sends a signal to the Raspberry Pi 5 indicating successful authentication. The green LED then lights up, and the LCD displays the message “Access Granted.” However, if no match is found, the Arduino sends a failed verification signal to the Raspberry Pi 5. In this case, the red LED turns on, and the LCD displays the message “Access Denied.” The process then returns to the initial state, allowing the user to attempt authentication again.

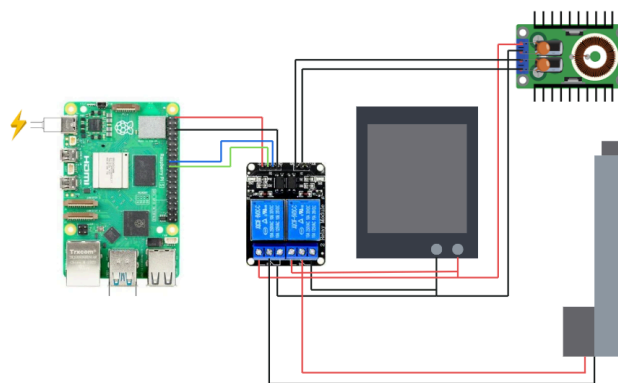
Furthermore, the system includes an automated feature that disables the fingerprint module whenever the ambient lighting is insufficient for optimal OpenCV performance. When this occurs, the LCD display shows the message “Too dark,” and a pop-up notification appears on the main Techno-Fitness display indicating “Fingerprint disabled.” This ensures that the user is immediately informed whenever lighting conditions may affect the accuracy and reliability of OpenCV’s visual detection process, thereby maintaining consistent system performance and precision.

For the second functionality, which is the camera elevation system, the team utilized a 12V linear actuator. The linear actuator provides the mechanical mechanism necessary to adjust the camera’s height in relation to the type of exercise being performed which involves the

extension and retraction of the linear actuator. This ensures that both standing-based and ground-based exercises are properly analyzed by the system. To implement this functionality, the linear actuator is connected to a separate 12V power supply since its operating voltage exceeds the power output capability of the Raspberry Pi 5. The actuator's operation is managed through a 2-channel relay module, which serves as a switch that allows the Raspberry Pi to control the actuator's motion without directly powering it. The relay module's input control pins (IN1 and IN2) are connected to GPIO pins 17 and 27 of the Raspberry Pi 5, while the relay's VCC and GND are powered from the Raspberry Pi's 5V and ground pins. The actuator's two motor terminals are connected to the relay's normally open (NO) and common (COM) terminals, allowing the relay to alternate the polarity of the actuator's input to extend or retract as needed.

In terms of operation, the actuator functions by converting electrical energy into linear motion. Inside the actuator, a small DC motor drives a lead screw that moves a rod in and out, producing an extending or retracting movement. The process begins when the Raspberry Pi 5 receives data from the exercise detection module. When the system detects that the user is performing a ground-based exercise, it sends a high signal from GPIO pin 17 to the relay, which activates the actuator and extends the rod to lower the camera for ground-level tracking. When the system identifies that the user has shifted to a standing-based exercise, the Raspberry Pi sends a signal through GPIO pin 27 to reverse the relay's switching direction, causing the actuator to retract and raise the camera to its elevated position. This automated adjustment allows the camera to continuously maintain an optimal viewpoint for OpenCV-based posture detection.

Figure 6
Actuator, Relay, and Monitor Circuit Diagram



The use of a linear actuator for camera elevation provides several key benefits. It ensures precise and stable vertical movement, allowing the camera to maintain consistent framing and image quality during exercise sessions. The actuator can handle heavier loads and maintain its position even when power is not continuously applied, ensuring that the camera remains steady during analysis. Furthermore, this automation enhances user convenience, as camera adjustments are performed seamlessly by the system based on exercise type, eliminating the need for manual repositioning. This ensures that OpenCV receives accurate visual data for motion tracking and posture analysis, contributing to the overall reliability and efficiency of the Techno-Fitness prototype.

For the third functionality, which is the light detection system, the team integrated a Light Dependent Resistor (LDR) to monitor the ambient lighting conditions surrounding the setup. The LDR serves as a sensor that determines whether the current light intensity is sufficient for OpenCV and MediaPipe to perform accurate motion tracking and posture recognition. This functionality ensures that the system operates under optimal lighting conditions, thereby maintaining accuracy and consistency in visual analysis. The LDR module is connected to the Arduino microcontroller via analog pin A5, where it continuously measures the ambient light intensity and transmits the data to the Raspberry Pi 5 for further processing.

In terms of operation, the process begins when the LDR detects the level of illumination in the environment. When light strikes the surface of the sensor, its resistance varies inversely with light intensity—high light levels result in low resistance, whereas low light levels produce high resistance. The Arduino interprets these changes as numerical values, which are then compared with a predefined threshold value. If the measured light intensity falls below this threshold, indicating inadequate illumination, the system automatically triggers a notification to alert the user that the environment is too dim for accurate OpenCV image processing. During this state, the fingerprint sensor is temporarily disabled to prevent false readings or recognition errors caused by insufficient lighting. Once the LDR detects that lighting has returned to acceptable levels, the system re-enables the fingerprint functionality and resumes normal operation.

The integration of the LDR provides several benefits to the Techno-Fitness system. It ensures that all visual components, particularly the OpenCV-based motion tracking module,

operate efficiently under proper illumination. This automated detection process minimizes posture analysis errors and enhances the overall reliability of the captured visual data. Moreover, it improves the user experience by providing real-time feedback on environmental conditions, allowing users to make necessary lighting adjustments before or during their workout sessions.

Testing and Evaluation Criteria

The Testing and Evaluation phase is a critical part of the development process, designed to ensure that the Techno-Fitness System performs effectively and meets its primary objectives. These objectives are focused on facilitating strength training and core training through computer vision-based posture assistance. The evaluation process includes a combination of technical tests and user-centered assessments to validate functionality, usability, and accuracy. The test would be conducted to validate the requirements of the objectives and to evaluate the system's efficiency and effectiveness into the detection of the posture using body landmarks to which would be judged based on the Raspberry Pi 5's operational limitations. The system will be evaluated based on the following core criteria:

1. **Posture detection** - The posture detection criterion assesses how accurately the system identifies and tracks the user's body keypoints, joint angles, and overall pose alignment during exercises. Using Mediapipe and OpenCV, the device must reliably recognize the user's body landmarks across various exercises such as squats, planks, etc. while differentiating between correct and incorrect postures.
2. **Real-time feedback performance** - This criterion measures the system's ability to process visuals and provide immediate, actionable feedback. The Techno-Fitness device must issue audio or visual cues within seconds of detecting improper form to prevent the continuation of incorrect movement.
3. **User accounts** - The system's fingerprint authentication feature will be evaluated for its ability to support multiple users with secure and distinct profiles. Each user account should store personal data such as exercise history, performance metrics, and progress scores. Through fingerprint authentication, the system should support multiple users with individualized records of performance.
4. **Usability of the interface** - This criterion focuses on the design and intuitiveness of the system's graphical interface. The interface must be visually clear, easy to navigate, and

responsive to real-time interactions during workouts. The interface must be simple to navigate and responsive enough to accommodate real-time workout adjustments.

5. **Integration Effectiveness** - This criterion evaluates how well all the hardware and software components function together as a unified system. The Raspberry Pi 5, camera module, sensors, fingerprint reader, and feedback interface must operate in sync without crashes, delays, or data loss. The mentioned must work consistently across repeated testing sessions.
6. **System Reliability** - System reliability measures the endurance and stability of the Techno-Fitness device during prolonged and repeated usage. The Raspberry Pi 5 must handle simultaneous processes such as camera streaming, real-time pose estimation, feedback generation, and data storage without overheating or system lag.

Refinement and Final Implementation Phase

After all key features were successfully tested and the exercise modules were fully integrated, the Refinement Phase began. This phase focused on optimizing the overall user experience while ensuring the system's stability, responsiveness, and reliability. The development team thoroughly reviewed all aspects of the Techno-Fitness System to identify and implement improvements in both functionality and presentation. During this phase, all bugs, glitches, and inconsistencies encountered during testing were resolved to ensure a smooth and seamless user experience. The user interface was significantly enhanced to become more intuitive, responsive, and visually appealing. Several quality-of-life improvements were introduced to simplify navigation, reduce user friction, and create a more engaging environment for performing workouts. These included refining visual cues, integrating multiple auditory feedback tunes, improving posture detection consistency, and enhancing layout responsiveness. Any technical issues identified during testing, such as software bugs, interface inconsistencies, or Mediapipe tracking errors, were thoroughly resolved. However, it was observed that adding additional computational conditions and processes to the already intensive software framework caused performance instability and occasional crashes on the Raspberry Pi 5. There was this one instance wherein a newly implemented condition (e.g. two second grace time in the dynamic exercise) wasn't able to work on the device as it completely crashed the device and the camera. It was resolved in a way to work hand in hand to the existing conditions, and optimized its code so that the Raspberry Pi 5 would not crash. This was primarily due to the limited processing

capacity and memory bandwidth of the hardware when executing concurrent image processing, pose estimation, and graphical interface tasks in real time.

User feedback gathered during the testing phase played a crucial role in the refinement process. Suggestions regarding comfort, accessibility, and usability were carefully reviewed and integrated into the final version of the system. Through this iterative improvement process, the final prototype was aligned with the needs and expectations of its intended users. Upon completion of all refinements, the Final Implementation Phase marked the full completion of the Techno-Fitness prototype, where the device is tested to a professional trainer and several respondents to provide their feedback and acceptance to the usage of the device. At this stage, all hardware and software components were integrated, optimized and validated based on prior testing and feedback. Documentation, such as evaluation forms and interviews, was prepared to support the final assessment of the system's performance. Demonstrations sessions were also conducted to familiarize users with the prototype's features.

List of Materials & Overall Cost

The hardware will serve as the backbone of the system, managing data collection, user interaction, and output display. The following table is the materials and components used in the completion of the prototype.

TABLE 3
Materials Used in the Prototype

Number of Component/s	Component Name	Amount Spent
1	Raspberry Pi 5 8GB Ram	Php 5,600.00
1	Raspberry Pi 5 Charger and 64GB SD Card	Php 524.22
1	Eizo NextScreen Monitor	Php 30,000.00
1	12V Linear Actuator	Php 1,768.00
1	Prototype Casing	Php 11,000.00
1	R305 Fingerprint Module	Php 2,500.00
1	BH1750 Ambient Light Sensor (I2C)	Php 45.00
1	Raspberry Pi 5 Kit (Casing and Cooler)	Php 604.00

1	Logitech Webcamera	Php 3543.00
1	Micro HDMI to HDMI	Php 243.00
N/A	Miscellaneous Materials (Wires, screws, etc.)	Php 300.00
1	Plastic Component Container	Php 249.00
1	Gyro Mouse and Keyboard	Php 243.00
1	12V - 15V Power Source	Php 220.00
1	2 Channel Relay Module	Php 76.00
1	3M Double Sided Tape	Php 169.00
N/A	Cable Clip and Zip Tie	Php 162.00
N/A	Couring Ball, Indoor Tape, and Corner Brace	Php 514.00
Overall Amount		Php 57, 760.22

Table 1 presents the list of hardware components integrated into the final Techno-Fitness device prototype, outlining the amount spent for each item required for the system. The design integrates computational, sensory, and mechanical hardware to support its core functionality: real-time fitness monitoring and feedback. At the core of the system is the Raspberry Pi 5 (8GB RAM), acting as the central processing unit, managing input/output and processing posture recognition. Power for the Pi and storage is provided by a Raspberry Pi 5 Charger and 64GB SD Card, and the Pi is housed in a Raspberry Pi 5 Kit including a casing and cooler. Visual output and the user interface are displayed on an Eizo NextScreen Monitor connected via a Micro HDMI to HDMI cable. User interaction is facilitated by a Gyro Mouse and Keyboard, and user authentication is handled by an Original Fingerprint Reader Module. Sensory input includes a Logitech Web camera for capturing user movements and a BH1750 Ambient Light Sensor to monitor room luminosity. The system incorporates mechanical movement via a Linear Actuator, controlled by a 2 Channel Relay Module which receives its power from a separate 12V - 15V Power Source. Structural components include a TV Stand, the main Thesis Casing, a Plastic Component Container, and various Miscellaneous Materials like wires, screws, 3M Double Sided Tape, Cable Clips, Zip Ties, Couring Ball, Indoor Tape, and Corner Braces for assembly and mounting. The overall amount spent on these components for the prototype totaled **Php 24,217.22**. This detailed hardware configuration ensures that the Techno-Fitness device can

deliver reliable, real-time feedback to users in a cost-effective, standalone solution suitable for beginner to intermediate fitness enthusiasts.

Figure 7
Concept Art and Physical Prototype



The figure above is a comparison between the proposed design and the physical prototype of the Techno-Fitness device. It illustrates the design of the Techno-Fitness System, showcasing a compact, ergonomic, and mobile fitness assistant tailored for home or indoor use. The device is mounted on a TV stand-like structure (central vertical stand with a stable, wheeled base) with a monitor frame mount, right-side panel, and an electronic container behind. A mounted camera is attached to an actuator, which serves as the core visual input for capturing the user's movements. Above the camera is a high definition monitor, which displays the user interface with an in-built speaker to output audio. Beside the monitor is the fingerprint and light sensor panel for the account management and light detection utilizing Arduino Uno. This design ensures that the system is both sturdy and portable, making it easy to position in different workout environments.

RESULTS AND DISCUSSION

The evaluation of the Techno-Fitness Prototype aimed to assess its functionality, performance, and user experience through both quantitative and qualitative approaches. Quantitative data were collected from twelve (12) participants who participated in the demonstration and testing of the system, while qualitative insights were obtained from an interview with a certified fitness coach. The results are systematically analyzed against the evaluation criteria established in the methodology, focusing on the system's ability to provide real-time posture assistance for calisthenics exercises using computer vision. These include *System Evaluation, Real-time Feedback Performance, User Accounts and Authentication, Usability of the Interface, Integration Effectiveness, System Reliability, and User Acceptance*.

System Evaluation

Following the presentation of results, this chapter discusses their implications, compares them against the project's objectives and related literature, and acknowledges the limitations encountered, ultimately assessing the effectiveness and potential of the developed prototype as a home-based fitness assistant. The primary purpose of the Techno-Fitness device is to function as a hardware-based fitness assistance system, integrating OpenCV with Mediapipe to offer users real-time workout posture feedback and assistance during home-based calisthenics workouts. This system aims to improve exercise consistency, promote safe workout habits by correcting form, and foster motivation, particularly for beginner to intermediate users. This chapter details the outcomes derived from the systematic testing and evaluation of the prototype, presenting the findings based on the specific evaluation goals and criteria previously established in the Materials and Methods section. Optimum conditions are observed during the testing of the prototype when monitoring the accuracy of the posture detection. It was evaluated for static, repetition-based, and dynamic exercises. Each exercise was tested by doing five (5) instances of a certain workout and a user attempting to do the correct posture for each workout then averaging the results of each instance to be plotted in the test result data tables below.

Table 4.1
Static Exercises Test Results

Exercise	Positives	False Positives	False Negatives	Positive %	Remarks
1. Cross Body Stretches	26.8	1.6	1.6	89.33	Excellent

2. Planking	23.2	3	3.8	77.33	Good
3. Superman Hold	23	4.2	2.8	76.67	Good
4. Standing Hamstring Stretch	28.8	0	1.2	96	Excellent
5. Cobra Stretches	26.8	1.2	2	89.33	Excellent
6. Standing Quad Stretches	25.4	1.2	3.4	84.67	Excellent
7. Leg Stretches	28.4	0.6	1	94.67	Excellent
AVERAGE	26.05	1.69	2.26	86.85	Excellent

0 - 19.99% = *Very Poor/Unacceptable*

20% - 39.99% = *Poor*

40% - 59.99% = *Fair/Moderate*

60 - 79.99% = *Good*

80% - 100% = *Excellent*

In the static exercise, the test was based on the duration meaning time. Each exercise was performed for thirty (30) seconds in five (5) instances. Positive is the duration to which the pose detection was correct. While the false positives/negatives are either misreads or inaccurate detections. Most exercises tested, including Cross Body Stretches (89.33%), Standing Hamstring Stretch (96%), Cobra Stretches (89.33%), Standing Quad Stretches (84.67%), and Leg Stretches (94.67%), achieved "Excellent" accuracy. This indicates the system consistently recognized the correct posture based on the 93% similarity threshold for the majority of the time these poses were held. Standing Hamstring Stretch notably had zero false positives, while Leg Stretches showed very few false negatives. Planking (77.33%) and Superman Hold (76.67%) registered slightly lower accuracies, falling into the "Good" category. The data suggests Planking was more prone to false negatives (average 3.8), meaning the system occasionally failed to recognize the correct pose even when held. Conversely, Superman Hold exhibited higher false positives (average 4.2), indicating instances where the system might have incorrectly registered the pose as correct during minor variations or transitions. Issues such as the exercise being floor based have contributed to it as partial occlusion of some body landmarks can cause the system to not detect the posture correctly. In addition to this, issues such as unstable Mediapipe landmark detection due to lighting or background being the same shade of color as the skin, occlusion, camera angle/perspective, or an unsuitable reference image are factors that are observed to have implications to the overall detection of the posture. It was also noticed that certain exercises such as the superman hold have a high false positive meaning that even if the user isn't completely

doing the correct posture as long as the angles are similar to that of the reference image. It would be detected as correct. Meanwhile other exercises such as planking have a higher false negative as even if the user were to do the correct posture. If there is partial occlusion of body landmarks, not fully in frame and a poor camera angle. It would detect the posture as wrong. Despite these minor variations, the overall average accuracy strongly suggests that the posture detection for static exercises using the 93% threshold is reliable and effective for the tested movements.

Table 4.2
Repetition Exercises Test Results

Exercise	Positives	False Positives	False Negatives	Positive %	Remarks
1. Jumping Jacks	10.6	1.2	0.2	88.33	Excellent
2. Body Weight Lunges	7	0.6	4.4	58.33	Moderate
3. Body Weight Squats	10.6	0.2	1.2	88.33	Excellent
4. Glute Bridges	9.7	1	1.2	81.67	Excellent
5. Bent Over Rows	8.4	0.2	3.4	70	Good
6. Leg Raises	9.8	2	0.2	81.67	Excellent
7. Push Ups	10.8	0.2	1	90	Excellent
8. Knee Push Ups	10.8	0	1.2	90	Excellent
AVERAGE	9.73	0.68	1.6	81.04	Excellent

0 - 19.99% = Very Poor/Unacceptable

20% - 39.99% = Poor

40% - 59.99% = Moderate

60 - 79.99% = Good

80% - 100% = Excellent

For the testing of the repetition-based workouts, each exercise was conducted twelve (12) repetitions in five (5) instances. With this, the system showed Excellent overall performance, achieving an average accuracy of 81.04%. The system was generally more prone to missing correct repetitions (false negatives, averaging 1.6 per trial) than incorrectly counting movements (false positives, averaging 0.68 per trial). Exercises involving distinct up/down phases like Leg Raises (90%), Push Ups (90%), Knee Push Ups (90%), Jumping Jacks (88.33%), Body Weight Squats (88.33%), and Glute Bridges (81.67%) all performed excellently, suggesting their motion rules effectively captured the two-state transitions with minimal missed reps. Performance was

'Good' for Bent Over Rows (70%) and dipped to 'Moderate' for Body Weight Lunges (58.33%), primarily due to significantly higher rates of false negatives (3.4 and 4.4, respectively).

The lower positive percentage for Bent Over Rows (70%) and particularly Body Weight Lunges (58.33%) primarily stems from a higher number of missed correct repetitions (false negatives). For Lunges, this is likely due to the complexity of coordinating both legs simultaneously, one bending deeply while the other remains straighter making it difficult for the system to consistently track both if one leg partially obscures the other. In the case of Bent Over Rows, potential variability in the user's torso angle during the movement can affect the relative wrist height measurement the system uses, making it harder to reliably distinguish the 'up' and 'down' phases. Additionally, for both exercises, partial occlusion of limbs or poor user camera perspective during the movement can prevent the system from gathering enough landmark data to confirm a complete repetition, thus failing to count valid reps. With this, it was identified that the detection for repetition-based workouts include visual input, partial occlusion of the user's body, incorrect user distance from the camera, or the presence of multiple people in the frame, can all degrade landmark detection.

In the context of repetition counting for exercises like push-ups within the Techno-Fitness system, false positives and false negatives represent critical metrics for evaluating detection accuracy. A false positive occurs when the system erroneously increments the repetition count despite the user not completing a full, valid repetition according to the defined motion rule. This can arise from partial movements, such as slight elbow bends during position adjustment, or from different exercises like bicep curls that might share similar joint angle changes (e.g., elbow flexion/extension) if the motion rule lacks sufficient specificity regarding the required range of motion, joint synchronization, or overall body posture. Conversely, a false negative occurs when the system fails to register a repetition that the user has correctly and fully performed. Such omissions might result from rapid execution exceeding the system's processing frame rate, temporary occlusion or poor visibility of key body landmarks preventing complete angle calculations. Both types of errors detract from the system's accuracy in tracking user performance.

Table 4.3
Dynamic Exercises Test Results

Exercise	Positives	False Positives	False Negatives	Positive %	Remarks
1. Arm Circles	26.6	0	3.4	88.67	Excellent
2. Hip Circles	29	0.8	0.2	96.67	Excellent
3. Shoulder Rolls	29	0.6	0.4	96.67	Excellent
OVERALL	28.2	0.47	1.33	90.35	Excellent

0 - 19.99% = *Very Poor/Unacceptable*

20% - 39.99% = *Poor*

40% - 59.99% = *Moderate*

60 - 79.99% = *Good*

80% - 100% = *Excellent*

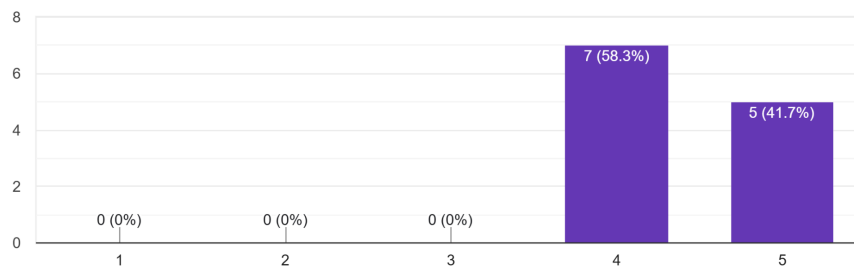
For the dynamic exercises, it is tested similarly to static exercises wherein instead of repetitions, it is based on duration (seconds). With this, the dynamic exercise detection performed with excellent overall accuracy, averaging 90.35% across the tested exercises. This high score suggests the system's approach of tracking continuous landmark movement against predefined motion patterns is effective for these types of activities. Hip Circles (96.67%) and Shoulder Rolls (96.67%) showed outstanding reliability with minimal errors, likely due to their localized and consistent rotational movements being easily captured by their specific motion rules. Arm Circles (88.67%), while still rated excellent, had a slightly higher rate of false negatives (3.4s average time missed). The system demonstrated the capability to identify these uninterrupted motions, for example, by computing the wrist's angular displacement relative to the shoulder for arm circles and comparing the rate and direction of change against expected values for a circular path. The two (2) second grace period helped in smoothing the function avoiding the abrupt changes from correct and wrong and if not implemented there would be higher false negatives/positives. However there are instances wherein a false positive is detected. It occurs when a user were to do a similar motion that meets the motion rule even if it is incorrect. In addition to this, larger and potentially faster movements increase the chance of brief landmark occlusion or deviations from the expected path, causing the system to momentarily lose track despite the two-second grace period intended to mitigate such issues. Overall, the low average false positive time (0.47s) indicates the motion rules are generally successful at distinguishing the correct dynamic exercises from incorrect movements or rest.

Quantitative Evaluation

Proceeding to the formal evaluation of the prototype, the device was tested by its target population, identified as beginner to intermediate fitness enthusiasts. The researchers adopted a quantitative approach as part of the final implementation phase. Following demonstration sessions and hands-on testing, evaluation forms were distributed to gather structured feedback. A total of twelve (12) respondents completed these evaluations. The following data summarizes their quantitative assessment of the prototype's functionality and user experience.

FIGURE 8.1
Device Ease of Use

The device is easy to set up and begin a workout session (e.g., login, positioning).
12 responses

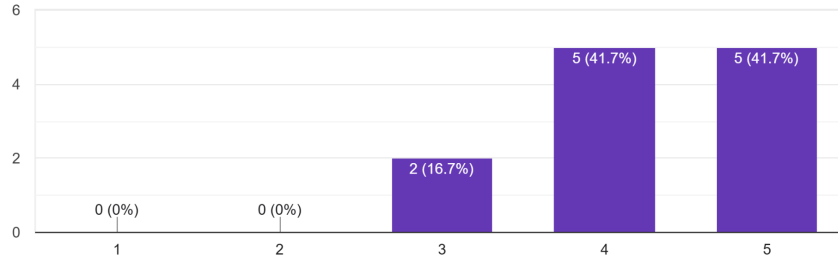


The survey results for device setup show an overwhelmingly positive response from the twelve (12) participants, with 100% of users rating the process favorably, a score of four (4) for "Agree" or five (5) for "Strongly Agree". Specifically, 58.3% or seven (7) users agreed, and 41.7% or five (5) users strongly agreed that the device is easy to set up and begin a workout session with, while no users rated the process as neutral or difficult. This high degree of usability likely stems from several key design choices. The system features a guided prerequisites screen that provides real-time feedback on lighting conditions and ensures the user is correctly positioned with at least 80% body visibility before the session starts. Furthermore, the biometric fingerprint recognition simplifies the login process, and the automatic camera elevation system removes the burden of manual adjustment by moving the actuator to the correct height for either standing or floor-based workouts.

Figure 8.2
Real-time Guidance Satisfaction

The real-time guidance system helped me perform the calisthenics exercises with proper and safer form.

12 responses

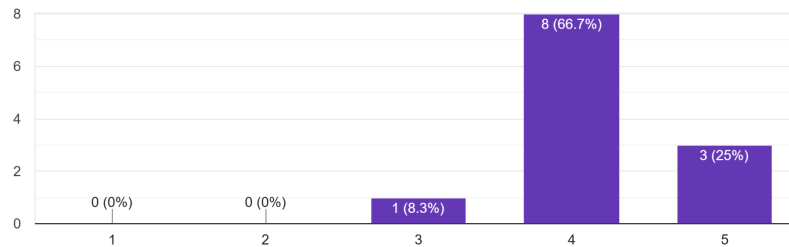


The survey results show a strong positive consensus on the system's core function, with a combined 83.4% of the twelve (12) users agreeing (41.7%) or strongly agreeing (41.7%) that the real-time guidance helped them perform exercises with proper and safer form. No users disagreed, though two (2) or 16.7% remained neutral. This positive outcome is a direct result of the system's feedback loop. By combining real-time visual cues (like the Mediapipe skeleton outline, matching percentages, and status icons) with auditory alerts (like tunes, beeps, etc.), the device notifies users of incorrect posture. For static exercises, the timer-stop mechanism effectively forces users to correct their form to meet the 93% threshold. For repetition-based and dynamic exercises, the system checks against specific motion rules to validate form before counting a rep or advancing the timer, reinforcing correct technique. The small neutral contingent (16.7%) may be due to the system's known limitations, such as feedback being non-verbal (beeps, not detailed voice commands) or occasional detection inaccuracies (false negatives) in more complex exercises.

Figure 8.3
Confidence in Exercises Satisfaction

I feel more confident in my exercises because the device acts as a reliable substitute for a professional coach.

12 responses

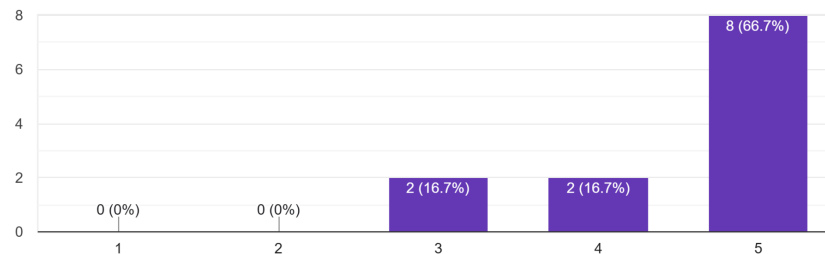


The survey results indicate that the device successfully builds user confidence, with a significant majority of 91.7% of respondents agreeing (66.7%) or strongly agreeing (25%) that the device acts as a reliable substitute for a professional coach. Only one (1) user or 8.3% was neutral, and no users disagreed, underscoring the system's effectiveness in this role. This high level of confidence is a direct result of the system's core design, which aims to replicate the immediate, corrective guidance of a human coach, a resource often inaccessible due to high cost. The device provides continuous, real-time posture analysis by comparing the user's form against validated motion rules and reference angles. When deviations are detected, the system provides immediate, actionable feedback through both visual (match percentages, icons, skeleton outlines) and auditory cues, effectively preventing incorrect repetitions and reinforcing proper, safer form. This function directly addresses the primary challenge of unguided home workouts, the high risk of performing movements incorrectly, and by doing so, successfully simulates a coach's watchful eye, thereby boosting the user's confidence in their own performance.

Figure 8.4
Real-time Visual Feedback Satisfaction

The real-time visual feedback (matching percentage, outline) clearly shows me if the posture being done is wrong or not.

12 responses

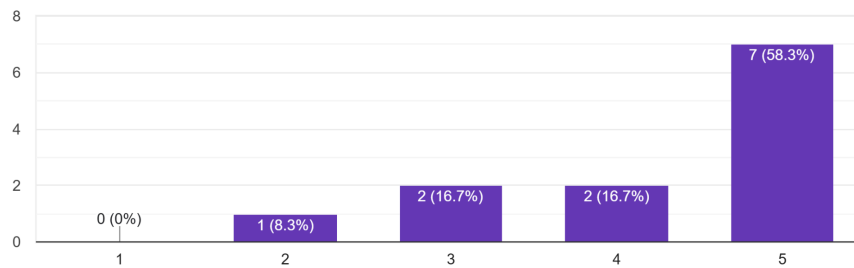


The survey data shows strong positive feedback on the clarity of the real-time visual cues, with a combined 83.4% of users agreeing. Two (2) or 16.7% agreed while eight or 66.7% strongly agreed that the feedback clearly showed if their posture was wrong. No users disagreed, though two (2) or 16.7% remained neutral. This positive reception is likely due to the multi-layered feedback system implemented in pose detection. The system provides an on-screen "outline" (the MediaPipe skeleton) drawn directly over the user's live video feed, allowing them to visually compare their form to the reference image or GIF. This is reinforced by explicit, color-coded indicators: a real-time matching percentage for static exercises (judged against the 93% threshold) and clear status icons (✓, -, X) for dynamic and repetition-based exercises to instantly signal correct, grace period, or incorrect movements. The small neutral percentage

(16.7%) might reflect minor system lag due to the Raspberry Pi 5's processing limitations, which causes a slight delay between the user's action and the visual feedback, making it feel less instantaneous.

Figure 8.5
Audio Cues/Alerts Satisfaction

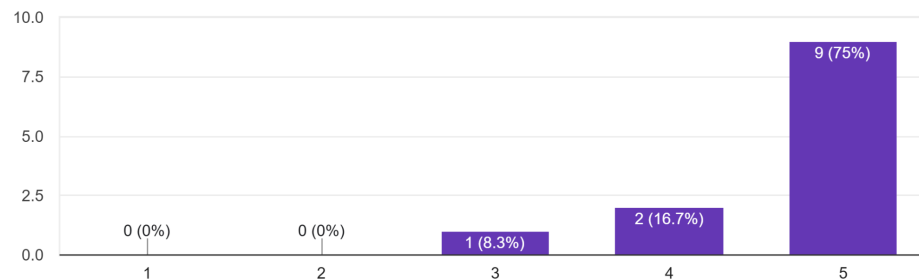
The audio alerts are effective and prompt me to immediately correct my posture.
12 responses



The survey results for audio alerts show a strong positive consensus, with 75% of users either agreeing (16.7%) or strongly agreeing (58.3%) that the alerts were effective and prompted immediate posture correction. Only one (1) user (8.3%) disagreed, and two (2) or 16.7% were neutral. This high level of effectiveness is likely due to the system's use of immediate, event-driven audio cues. The pose detection function uses pygame.mixer to play distinct sounds for specific events, most notably the wrong tune when an incorrect posture is held too long, a beep for a good rep, and text-to-speech voiceline if the user is out of frame or a workout is floor based. This auditory feedback works in conjunction with the visual indicators (e.g., red 'X') to alert the user instantly, allowing them to make corrections without needing to constantly stare at the screen. The small neutral/disagreeing contingent might be attributed to the nature of the feedback wherein beeps/tunes are played rather than specific voice commands to which are stated in the scope and limitations or minor system latencies under heavy processing load.

Figure 8.6
Biometrics/Login Satisfaction

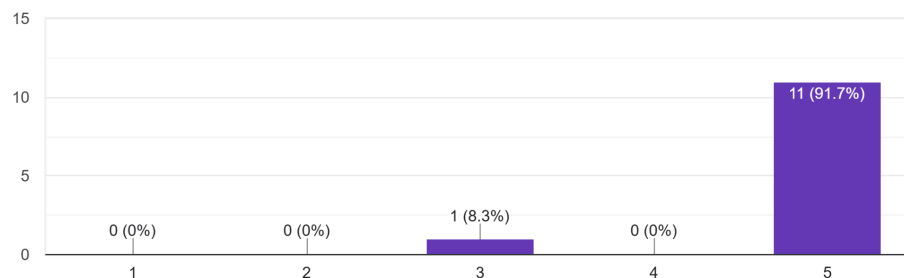
I find the fingerprint login to be reliable and secure.
12 responses



The survey data shows a very strong positive response regarding the fingerprint login, with 91.7% of the twelve (12) users agreeing or strongly agreeing that it is reliable and secure. This is dominated by 75% or nine (9) users who "Strongly Agreed" with a score of five (5), supplemented by 16.7% or two (2) users who "Agreed" with a score of four (4). Only one (1) user (8.3%) was neutral, and critically, zero users reported a negative experience. This high level of confidence is likely due to the implementation of a dedicated biometric authentication system. The system uses a specific Fingerprint Reader Module which is handled by a separate Arduino Uno microcontroller. Offloading this task from the main Raspberry Pi 5 to a dedicated controller ensures that the login process is consistent, fast, and not affected by the main application's processing load. Furthermore, biometrics are inherently perceived by users as more secure than traditional passwords, and the system's ability to reliably load distinct user accounts reinforces this sense of both reliability and security.

Figure 8.7
Interface and Navigation Satisfaction

The overall user interface and navigation are intuitive and simple to use.
12 responses

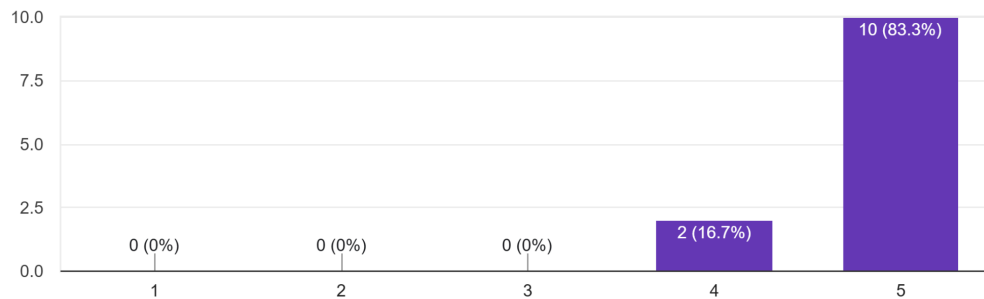


The survey results show exceptionally high approval for the system's user interface and navigation, with 91.7% eleven (11) of the twelve (12) 'Strongly Agreeing' that it is intuitive and simple to use. Only one (1) user (8.3%) remained neutral, and no users reported any difficulty. This strong positive feedback is likely due to the clear, process-oriented design of the application, which is built on PyQt6. The system guides the user through a logical, step-by-step flow, as defined in the development flowcharts: from a simple biometric login, to the main menu with a clear calendar interface, and then into the guided prerequisite checks before the workout begins. During the workout, pose detection interface provides constant, clear feedback such as live camera feeds, match percentages, and rep counters making the interaction intuitive as the user immediately sees the effect of their actions. Furthermore, the reliance on a simple gyro

mouse and keyboard for navigation, rather than more complex controls, likely contributes to this perceived simplicity and ease of use.

Figure 8.8
Camera Elevation Satisfaction

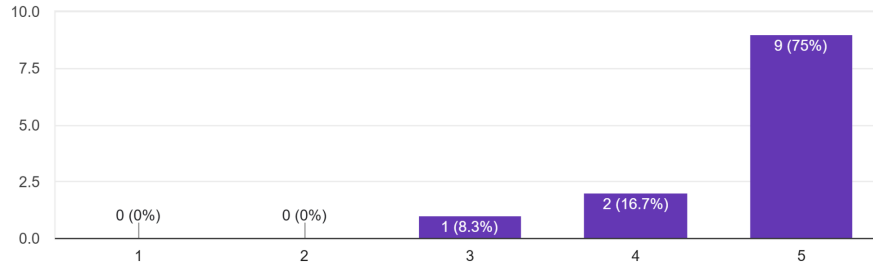
The camera elevation feature effectively and easily adjusts to different types of exercises (e.g., standing vs. floor exercises)
12 responses



The survey data for the camera elevation feature indicates exceptional user satisfaction, with 100% of the twelve (12) respondents agreeing or strongly agreeing that it adjusts effectively and easily. An overwhelming majority, 83.3% or ten (10) users, strongly agreed with a score of 5, while the remaining 16.7% or two (2) users agreed with a rating of four (4). This resoundingly positive result is likely due to the feature's adjustment if the workout is either floor-based or standing to which requires zero effort from the user. Based on the system's design, the pose detection function automatically checks a flag (e.g., "floor_based": true) in the workout plan's JSON file before an exercise begins. The system then sends a command to the 2-channel relay, which in turn drives the linear actuator. This mechanism automatically moves the camera to the correct height for either standing or floor-based exercises without any user intervention. This automation completely eliminates the most common point of friction for such systems, manually repositioning the camera, making the transition between different exercise types both "effective" and "easy" from the user's perspective.

Figure 8.9
Prerequisites Satisfaction

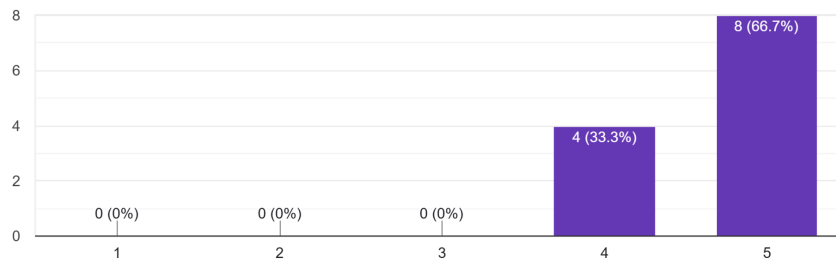
The prerequisite check (light/visibility) is helpful and ensures the system works accurately.
12 responses



The survey data indicates strong approval for the prerequisite check feature, with 91.7% of the 12 users agreeing or strongly agreeing that the light and visibility check is helpful and ensures accuracy. A significant majority, 75% or nine (9) users, strongly agreed with a score of five (5), while 16.7% or two (2) users agreed with a score of (4). Only a single user (8.3%) was neutral, and no users disagreed. This positive reception is likely because this feature directly addresses the system's known operational limitations. Instead of allowing a user to start a workout in suboptimal conditions (like a dark room or poor positioning) which would lead to frustratingly inaccurate detection and missed repetitions, the Prerequisites Tab acts as a crucial "safety check". By prompting the user to meet these requirements before the session begins such as ensuring sufficient lighting (monitored by the BH1750 light sensor) and verifying that at least 80% of body landmarks are visible, the system guarantees it is operating within the optimal conditions required by Mediapipe and OpenCV. This proactive step ensures the subsequent pose detection is reliable and accurate, which the users clearly recognized as helpful.

Figure 8.10
User Engagement and Experience

I feel engaged and motivated during a workout routine.
12 responses

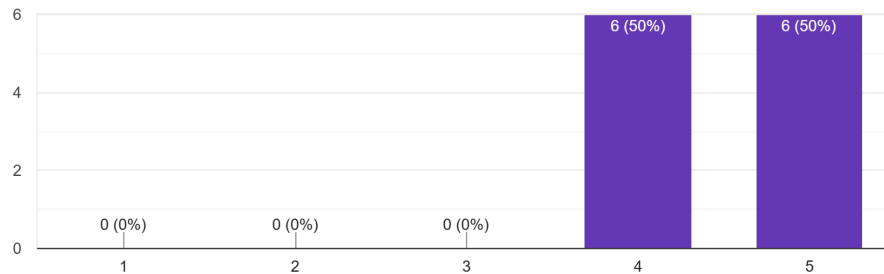


The survey results for user engagement and motivation are exceptionally strong, with 100% of the 12 respondents agreeing or strongly agreeing that they felt engaged and motivated

during the workout routine. This positive feedback is dominated by a 66.7% or eight (8) users majority who "Strongly Agreed" with a score of five (5), with the remaining 33.3% or four (4) users "Agreeing" with a score of four (4). No users reported a neutral or negative experience. This unanimous feedback indicates the system's design successfully addresses key motivational challenges. The real-time, interactive feedback loop which includes live posture outlines, matching percentages for static poses, and immediate rep counting for dynamic/rep exercises effectively gamifies the workout experience. By providing constant, clear metrics of success and failure, the system turns the exercise into an engaging challenge rather than a passive routine. This aligns with the project's theoretical framework by fulfilling the user's need for competence and providing clear prompts and ability-enhancing feedback, ultimately fostering the motivation and consistency the project set out to achieve. One user captured this sentiment by describing the experience as "very much engaging and fun," attributing this positive feeling to the novel experience of "doing exercises with something that guides/assists you throughout the session." This interactive guidance, the respondent noted, directly "makes one motivated to complete the exercise". This feedback confirms the system's effectiveness in transforming a potentially monotonous routine into an interactive challenge, aligning with the project's objective to enhance user engagement and motivation through real-time feedback.

Figure 8.11
Techno-Fitness Motivation Evaluation

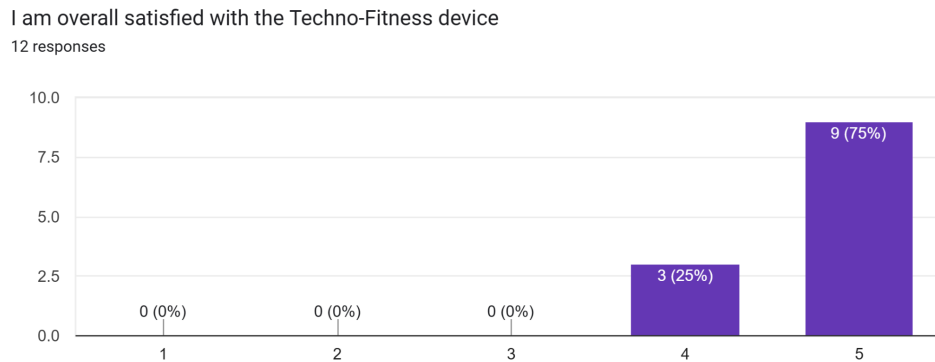
Using the Techno-Fitness Device motivates me to maintain consistency in my workout routine.
12 responses



The survey data shows a unanimous positive response regarding motivation and consistency, with 100% of the twelve (12) users indicating the device helped them maintain their workout routine. The responses were evenly split, with 50% or six (6) users selecting "Agree" with a score of four (4) and 50% or (6) users selecting "Strongly Agree" with a score of five (5). This result strongly suggests the device successfully addresses the documented user struggles

with consistency. The motivation likely stems from the adaptive scheduling and progress tracking features. The system provides a clear, validated workout plan for each day, removing the "lack of knowledge" barrier. More importantly, the application tracks user adherence through last workout date and completed dates parameters. The adaptive scheduling logic enforces rules that create accountability, such as repeating a week if 3 days are skipped or resetting the schedule after 5 consecutive missed days. This structure, combined with gamification elements like real-time performance scores and visual progress tracking, acts as a powerful prompt and reward system, aligning with the Fogg Behavior Model to motivate users to return consistently.

Figure 8.12
Overall User Satisfaction



As presented in figure num, all respondents are reported to have a good satisfaction rating with the Techno-Fitness prototype with nine (9) or 75% of the respondents indicating *strong agreement* (Excellent) and three (3) or 25% of the respondents expressing *agreement* with the statement, “*I am overall satisfied with the Techno-Fitness Device.*” This unanimous positive response demonstrates that the system successfully met user expectations in terms of performance, usability, and effectiveness. The high satisfaction level suggests that users found the prototype engaging and reliable in assisting them during workouts. Moreover, the results imply that the integration of real-time feedback and interactive features contributed to an enhanced fitness experience, leading to greater user approval and acceptance of the system.

Qualitative Evaluation

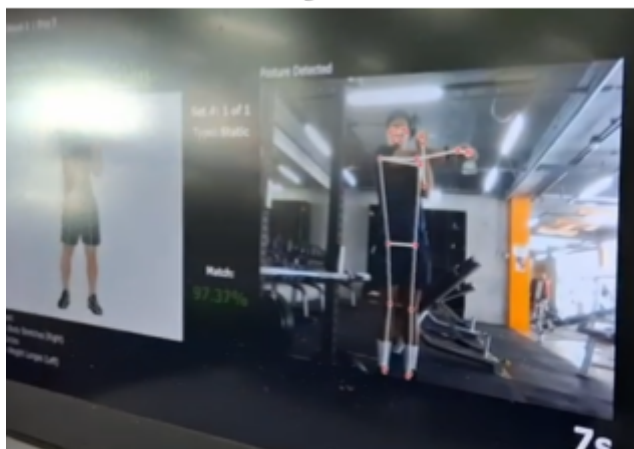
Proceeding onto the qualitative evaluation of the prototype. The evaluation of the professional *Techno-Fitness* prototype was conducted through a testing session in Outlets Lipa with a certified Fitness Coach, Engr. Michael Guanterro who served as the system evaluator. The primary goal of this evaluation was to assess the system's usability, functionality and practical applicability in real-world fitness environments. The evaluator

provided valuable insights that included both commendations for the system's strengths and constructive suggestions for its improvement. The evaluator recognized several strengths in the *Techno-Fitness* prototype that aligned with its intended objectives of improving workout monitoring and promoting proper exercise performance. One of the most notable points of positive feedback was directed towards the system's biometric fingerprint login. The evaluator particularly praised the biometric authentication system, describing the fingerprint login as significantly more convenient and secure than the team's initial RFID-based proposal, especially for a home environment with multiple users. Furthermore, the evaluator commented on the robustness of the motion detection, noting that the system sometimes could still correctly track his form even when other people were moving in the background. This observation is

FIGURE 9
Testing of Prototype with Fitness Coach



Figure 10
Fitness Coach Doing Stretches Static Exercises



noteworthy as there are instances wherein the detection would be confused because there are multiple people within the frame. This successful test under real-world conditions corroborates the fact that the pose detection pipeline, using OpenCV and Mediapipe, is highly effective and well-suited for a typical home environment. He also highlighted the effectiveness of the integrated grading and scheduling systems as key motivators. The

coach noted that the grading feature, which calculates a score based on performance metrics like inaccuracies and skipped exercises, provides a clear and essential incentive for users to focus on proper posture and correct execution. This, combined with the adaptive scheduling system that enforces consistency, for instance, by repeating a week if three or more days are skipped, creates a structured, guided experience. The coach concluded that this dual approach is particularly beneficial for the target beginner audience, who often lack the foundational knowledge for correct form and the self-discipline to maintain a consistent routine without such external guidance. He also liked the automated camera movement that easily adjusted its angle for following along as the user would do push-ups, thereby enhancing user experience and interaction. However, the evaluator pointed out an important concern with what he called the “Push-up Problem.”. When people are incapable of doing a regular push-up, they either try to do the movement repeatedly or skip the movement, which greatly limits learning and progression. This design, he declared, is not helpful in developing form and strength over time. In this light, he suggested to implement knee push ups before one goes to a full push-up. This was later-on implemented into the workout plan. This will help users of different proficiency levels develop progressive strength, making the system more inclusive and effective for learning.

Summary

In summary, the evaluation data comprehensively validates the Techno-Fitness prototype against all established criteria. The core System Evaluation of posture detection was quantitatively confirmed, achieving "Excellent" average accuracy for static (86.85%), dynamic (90.35%), and repetition-based (81.04%) exercises. This technical success translated directly into exceptionally high User Acceptance, with majority of the participants feeling engaged, motivated and expressing overall satisfaction. This positive user response is supported by the high ratings for Real-time Feedback Performance, where 83.4% found visual cues clear and 75% found audio alerts effective, leading 91.7% to feel more confident in their form. The Usability of the Interface was rated near-perfectly, with users finding the setup and overall navigation (91.7% strongly agreeing) intuitive. This seamless experience is a testament to the Integration Effectiveness of the system's components, demonstrated by the majority of the users having high satisfaction with the automated camera elevation and 91.7% approval for the helpfulness of the prerequisite checks. User Accounts and Authentication were also confirmed as successful, with 91.7% of

users and the expert evaluator finding the biometric login reliable and secure. Finally, System Reliability was demonstrated by the consistent high-accuracy scores across multiple test instances and the prototype's ability to function robustly in real-world scenarios, even while acknowledging the Raspberry Pi 5's inherent processing limitations.

CONCLUSION AND RECOMMENDATION

The increasing demand for accessible home fitness solutions, coupled with the limitations of existing applications in providing real-time form correction, established the need for the Techno-Fitness device. This project aimed to develop and evaluate a prototype utilizing computer vision specifically OpenCV and Mediapipe on a Raspberry Pi 5 to monitor, analyze, and guide users through calisthenics exercises with correct posture. The research successfully produced a functional device capable of detecting body landmarks in real-time and evaluating user form against predefined references and motion rules for static, repetition-based, and dynamic exercises. This project successfully achieved its general objective: to develop a functional prototype of a device that utilizes real-time body keypoint detection through OpenCV and Mediapipe to monitor, analyze, and correct a user's posture during physical exercises. The research produced a viable, standalone system built on the Raspberry Pi 5, capable of evaluating user form against predefined rules for static, repetition-based, and dynamic exercises. With this, the project also met its specific objectives:

- The posture detection capabilities of the Mediapipe API were explored and validated, first by implementing simple static exercises and calibrating a 93% similarity threshold, which proved to be a robust benchmark.
- Posture detection feedback was applied to a full calisthenics workout plan, with the Raspberry Pi 5 (acting as the central microcontroller) processing landmarks to deliver immediate visual and auditory cues.
- A feature was implemented to compare user posture against predefined references, using angle calculations for static types and custom-coded motion rules for dynamic and repetition types.

- A biometric authentication system using a fingerprint module and an Arduino interface was successfully integrated, providing secure, multi-user access. User evaluation confirmed this feature was reliable and secure.
- A camera elevation system using a linear actuator was integrated, effectively adjusting the camera's height to accommodate both standing and ground-based workouts, a feature users found highly effective.
- A light sensor was integrated via the Arduino to ensure the API's accuracy, successfully implementing a prerequisite check that users found helpful.
- The system's efficiency and effectiveness were evaluated, demonstrating high accuracy in quantitative tests (86.85% for static, 81.04% for reps, 90.35% for dynamic exercises) and receiving strong positive user acceptance for its role in improving form and motivation.

In conclusion, the Techno-Fitness device presents an innovative, viable, and efficient solution that bridges the gap between cost-prohibitive personal trainers and non-responsive fitness apps. The device successfully assists beginner to intermediate users in improving workout consistency and promoting safe exercise habits, fulfilling its core purpose.

Recommendations

Based on the findings and identified limitations, future development should focus on several key areas for improvement to enhance the capabilities and user experience of the Techno-Fitness device. First, feedback modalities should be significantly enhanced. The current system's reliance on simple audio cues (beeps and tunes) was noted as a limitation. Integrating a text-to-speech (TTS) engine would provide specific, corrective verbal instructions (e.g., "Raise your hips" or "Straighten your back"), more closely emulating the real-time guidance of a professional coach and addressing feedback from evaluators. To improve detection robustness, future work should explore the development of custom machine learning models trained on diverse datasets. This would help overcome MediaPipe's current sensitivities to suboptimal lighting, varied camera perspectives, and partial body occlusions. As noted in the findings, implementing more complex logic already strained the Raspberry Pi 5, causing performance instability. Therefore, deploying more advanced custom models would necessitate migrating the core processing to a more powerful processing unit than the Raspberry Pi 5, such as a

single-board computer with a dedicated AI accelerator, to maintain real-time performance. Addressing the reliance on fixed routines, AI-driven personalization could be implemented. This would evolve the existing rule-based adaptive scheduler (which repeats weeks based on skips or grades) into a truly dynamic system capable of generating or modifying daily workout plans based on the user's specific performance metrics, goals, and long-term progress. To overcome the constraints of the current purely offline system, a companion mobile application should be developed. This would enable cloud synchronization, allowing users to access their workout history, track long-term data, and manage their profile remotely, thereby addressing the data access limitations noted in the study. Finally, the camera elevation system could be evolved from its current preset-based mechanism into an active, dynamic user tracker. By leveraging the existing landmark data from MediaPipe, the system could be programmed to automatically pan and tilt, keeping the user centered in the frame during more complex or mobile exercises.

Conclusion

The Techno-Fitness device successfully demonstrated its value as an innovative and effective solution to the problem of unguided home-based calisthenics. By integrating OpenCV and Mediapipe on a Raspberry Pi 5, the prototype met its primary objective of providing real-time feedback on user posture across a validated workout plan. Quantitative testing confirmed the system's high accuracy in evaluating static, repetition, and dynamic exercises, while qualitative user feedback was overwhelmingly positive. Participants found the system's guided positioning, automated camera elevation, and biometric login features to be intuitive and reliable. Most importantly, users felt the real-time feedback loop made them more engaged, confident, and motivated to maintain consistency and proper form, validating the device as a reliable and cost-effective substitute for a professional coach. Overall, this study concludes that the Techno-Fitness prototype presents a viable and efficient tool for assisting beginner to intermediate fitness enthusiasts, successfully bridging the gap between digital convenience and the necessity of safe, guided exercise.

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APPENDICES

Letter of Intent

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April 25, 2025

Dear Engr. Michael Guantero,

Greetings of Peace and Joy!

We are undergraduate students of **BS Computer Engineering** from **De La Salle Lipa**, currently working on our thesis project entitled "**Techno-Fitness: The Mobile Posture and Form Correction Coach.**" This study aims to develop an AI-powered fitness assistant that enhances exercise accuracy and consistency through real-time feedback, computer vision, and pose detection technologies.

As part of our research process, we are gathering data to identify essential exercises, develop an effective workout plan, and obtain other necessary inputs to help conceptualize and enhance the functionality of our thesis project. We humbly request your participation in our survey/interview to help us gain valuable insights that will shape the development of this innovation. Your input will be treated with utmost confidentiality and used solely for academic purposes.

We assure you that this will only take a few minutes of your time and your honest responses will significantly contribute to the success and impact of our research.

Thank you very much for your time and support.

Sincerely,

Karl Ambrose Latayan
Thesis Leader

May we kindly ask for your consent to proceed with the recording and the interview?

Table 4
Interview Guide Questions

Basic Information	• What is your name? • What is your age? • What is your occupation or field of work/study? • How would you describe your current level of physical activity or fitness routine?
Experience with Fitness Technology	• Have you used any fitness applications or devices before (e.g., smartwatches, fitness apps, motion sensors)? - <i>Follow-up:</i> Which ones, and what features did you find most helpful or lacking? • Do you currently use any AI-driven or interactive fitness platforms? - <i>Follow-up:</i> How would you describe your experience with them?
Perception of the Techno-Fitness System	• What is your initial impression of the proposed Techno-Fitness System? • Do you think real-time feedback on form and posture would be useful during your workouts? Why or why not? • How important is personalization (e.g., adaptive workout plans) to your fitness journey? • Would a system like this increase your motivation or commitment to exercise regularly?
Usability and Design Feedback	• What features would you expect or desire in such a fitness system? What are the most effective exercises for someone with limited equipment or space?



	• What concerns or limitations do you foresee with an AI-powered fitness assistant? • Would you be comfortable using a system that visually analyzes your body posture through a camera? - <i>Follow-up:</i> What privacy or security concerns might you have?
Closing	• Do you have any suggestions for how this system could be improved or made more engaging? • Is there anything else you'd like to add regarding your experience with technology in fitness?

Survey Questions

Techno-Fitness: The Posture and Form Correction Coach Survey

Greetings of Peace!

We are 3rd year students from the Bachelor of Science in Computer Engineering program, currently conducting a study on an AI-powered fitness device called the Techno-Fitness: The Posture and Form Correction Coach Device. Techno-Fitness is an AI-powered fitness system designed to guide users through proper workout routines by providing real-time corrective feedback, exercise modes, and motivational support. It combines computer vision and machine learning to monitor user posture and movements, helping improve workout form, prevent injuries, and promote consistent, engaging, and effective fitness habits especially for individuals without access to professional trainers.

We would greatly appreciate your time in answering this short survey to help us evaluate the system's effectiveness and improve its features.

Data Privacy Act

We value and respect your privacy in accordance with the Data Privacy Act of 2012 and all related policies. By proceeding with this form, you are voluntarily giving your informed consent for the researchers to collect and use the data solely for the purpose of this study. All



information gathered will be used strictly for the enhancement and further development of the Techno-Fitness Device.

Your responses will remain completely anonymous and confidential. Kindly read each question carefully and answer honestly based on the provided instructions. Your participation is greatly appreciated.

Table 5.1
Respondent's Profile Questions

A. Respondent's Profile	Choices
What is your age?	N/A
What is your sex?	Male/Female
How would you gauge your fitness level?	Beginner, Intermediate, and Advanced

Table 5.2
Participants' fitness habits, challenges, and interest Questions

B. Participants' fitness habits, challenges, and interest	1	2	3	4	5
1. I usually engage in fitness.					
2. I am familiar with various types of workouts and their purposes.					
3. I consistently engage in my daily responsibilities (e.g., studies, work, chores, etc.).					
4. I find fitness applications to be effective.					
5. I find it difficult to be consistent in					



engaging in fitness.					
6. I struggle with doing the correct form/posture in my exercises.					
7. Would you be interested in an AI powered fitness device that provides real-time feedback on your movements and form?					

1 - Highly Disagree; 2 - Disagree; 3 - Neutral/Neither Disagree nor Agree; 4 - Agree; 5 - Highly Agree

Acceptance Survey

Techno-Fitness: Leveraging Computer Vision and Workout Assistance Coach Evaluation

Greetings of Peace!

We are 4th year students from the Bachelor of Science in Computer Engineering program, currently conducting a study on an AI-powered fitness device called the Techno-Fitness: Leveraging Computer Vision and Workout Assistance Coach.

Techno-Fitness is a OpenCV (computer vision) powered device that utilizes Mediapipe for the detection of exercise form designed to improve exercise consistency, provide workout assistance, and engage the user in doing exercises. It is a fitness system designed to guide users through proper workout routines by providing real-time corrective feedback, performance tracker, and motivational support. In addition, it is used to monitor user posture and movements, helping improve workout form, prevent injuries, and promote consistent, engaging, and effective fitness habits especially for individuals without access to professional trainers.

We would greatly appreciate your time in answering this short survey to help us evaluate the system's effectiveness and improve its features.

Data Privacy Act

We value and respect your privacy in accordance with the Data Privacy Act of 2012 and all related policies. By proceeding with this form, you are voluntarily giving your informed consent for the researchers to collect and use the data solely for the purpose of this study. All information gathered will be used strictly for the enhancement and further development of the Techno-Fitness Device.

Your responses will remain completely anonymous and confidential. Kindly read each question carefully and answer honestly based on the provided instructions. Your participation is greatly appreciated.

Table 6.1
Respondent's Acceptance to Hardware Functionality

A. Hardware Component Functionality	1	2	3	4	5
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1. The device is easy to set up and begin a workout session (e.g., login, positioning).					
2. The camera elevation feature effectively and easily adjusts to different types of exercises (e.g., standing vs. floor exercises)					
3. I find the fingerprint login to be reliable and secure.					
4. The prerequisite check (light/visibility) is helpful and ensures the system works accurately.					
5. The overall user interface and navigation are intuitive and simple to use.					

Table 6.2
Respondent's Acceptance to Effectiveness and Assistance

B. Effectiveness and Assistance	1	2	3	4	5
1. The real-time visual feedback (matching percentage, outline) clearly shows me if the posture being done is wrong or not.					
2. The audio alerts are effective and prompt me to immediately correct my posture.					
3. The real-time guidance system helped me perform the calisthenics exercises with proper and safer form.					
4. I feel engaged and motivated during a					



workout routine.					
5. I feel more confident in my exercises because the device acts as a reliable substitute for a professional coach.					
6. Using the Techno-Fitness Device motivates me to maintain consistency in my workout routine.					
7. I am overall satisfied with the Techno-Fitness device					

Techno-Fitness Software Link:

https://drive.google.com/file/d/1HIN3r4yjNhoxCe45W1z54t7ZG_ftnoe8/view?usp=sharing

Techno-Fitness Flowchart Diagram HD Ver.:

<https://drive.google.com/file/d/1rH1LfdnWnzUyghDiBG50K1B0oiF9UGTN/view?usp=sharing>